

Do Informed Consumers Pay Less?

Evidence from a Survey with Linked Grocery Purchase Data*

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Abstract

This paper examines how consumer price knowledge affects shopping behavior and the prices consumers pay in grocery markets. We combine survey-based price recall data from over 2000 Norwegian households — yielding over 70 000 price recalls across two grocery chains and 24 products—with 18 months of linked individual-level transaction histories. Better-informed consumers—those who recall prices more accurately—pay lower prices by timing purchases to coincide with sales. A 10 percentage point increase in price knowledge (approximately the interquartile range) is associated with a 1.3 percentage point reduction in prices paid. Our results provide direct support for the central mechanism in Varian’s (1980) model of sales: that informed consumers pay lower prices by exploiting temporary discounts. We also find that consumers who are more active in seeking information about prices have higher price knowledge. Taken together, and with the caveat that we are only considering consumer responses here, our results suggest that policies or tools that help consumers learn about prices may be effective in enhancing competition. Our findings also speak to a marketing literature that seeks to measure and explain consumer price knowledge. By linking survey data to actual shopping behavior, we contribute to this literature by demonstrating that shopping behavior and attitudes are stronger predictors of price knowledge than demographic characteristics.

Keywords: Price Knowledge, Price learning effort, Price Competition, Sales Utilization, Search costs

JEL Codes: D83, L10, L66

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1 Introduction

Consumers’ knowledge of prices across retailers is crucial for effective market competition.¹ Acquiring price information requires effort, as consumers must allocate attention and often engage in various forms of search. The prevailing intuition – both among economists and policymakers – is that reducing search costs leads to more informed consumers, thereby intensifying competition and driving prices down.

A central modeling approach in the literature on costly search is to distinguish between two types of consumers—one with zero search costs (“informed”) and one with some positive search cost (“uninformed”), see e.g., [Salop and Stiglitz \(1977\)](#), [Varian \(1980\)](#) and [Stahl \(1989\)](#).² These models predict that a higher share of informed consumers intensifies competition and drives prices down—a result that has become central to economic research on search costs and market outcomes. Such models have for instance been used to generate predictions for the effects of mandatory price disclosure on grocery markets ([Ater and Rigbi, 2023](#)), price dispersion in gasoline retailing ([Pennerstorfer et al., 2020](#)), the role of temporary sales in grocery markets ([Villas-Boas, 1995](#)), and the impact of reduced search costs as consumers gain access to the internet ([Brown and Goolsbee, 2002](#)). They have also been tested in laboratory settings to better understand price dispersion and consumer behavior ([Morgan et al., 2006](#)).

While theory predicts that lower search costs lead to more informed consumers and greater price competition, empirical evidence on how price knowledge affects actual consumer outcomes remains limited. To this end, this is the first paper to provide direct evidence on how consumer price knowledge affects prices paid in the grocery market. Specifically, we ask: Do more informed consumers systematically pay lower prices? Using an online survey, we measure consumers’ knowledge of grocery prices and link their responses to 18 months of individual-level purchasing history from Norway’s largest grocery retail group. This unique combination of survey data and purchasing histories allows us to test the prediction that informed consumers pay lower prices, and to investigate whether price knowledge is associated with certain shopping behaviors, attitudes, or demographics. The final sample includes more than 2000 households and over 70 000 recalled prices across two grocery chains and 24 central grocery products.

Theoretical models such as [Varian \(1980\)](#) predict that when consumers become more informed, firms have stronger incentives to lower prices through temporary discounts and promotions to attract price-sensitive shoppers. A key assumption behind this prediction

¹[Stigler \(1961\)](#) is a key reference for the literature that has followed.

²While the intuition that lower search costs intensifies competition is attractive, models that feature this outcome are surprisingly intricate. Indeed, the formal literature of the interaction between search costs and competition started with a paradox—the result in [Diamond \(1971\)](#) that introducing even minuscule search costs into a model of Bertrand competition with homogeneous products can allow all firms to charge monopoly prices instead of pricing at marginal cost.

is that informed consumers exploit price variation by timing purchases to coincide with sales rather than paying regular prices. We provide direct empirical support for this mechanism: consumers with higher price knowledge systematically pay lower prices, not because they buy inherently cheaper products or shop at inherently cheaper stores, but because they are more likely to purchase products when the price is low. This suggests that informed consumers impose a competitive constraint on retailers by responding to price fluctuations, reinforcing the theoretical link between consumer information and price competition. The effects we find are economically significant: For instance, moving from quite low price knowledge (a difference between price recall and actual prices of 40%) to quite high price knowledge (a difference between price recall and actual prices of 10%) is associated with a 4 percentage-point decrease in the average price paid. This relationship holds even after controlling for demographic characteristics, shopping behavior, and stated price consciousness.

Having established the relationship between price knowledge and prices paid, we go on to investigate the drivers of price knowledge. We find that, as a whole, product characteristics and observed consumer behaviors are more important than demographic characteristics and other self-reported variables in explaining the price knowledge of the consumers. For example, a high number of substitutes strongly predicts lower price recall, while prior purchasing experience and spending on a product are associated with higher recall. However, some self-reported factors still matter. Women tend to have better price knowledge than men, for instance. We also find that consumers who are more active in gathering information about prices—by comparing prices in stores, reading flyers, or consulting price comparison surveys in the media—have higher price knowledge. This association also holds after controlling for differences in demographics, shopping behavior and the consumers’ reported price sensitivity, that is, how important prices are in the choice of products and stores.

Our findings have implications for policy design. First, the relationship between price knowledge and the prices consumers pay suggests that policies that successfully raise consumers’ price knowledge can lead to lower prices and stronger competition. Second, we find that consumers who actively engage in price-gathering activities tend to be more knowledgeable. This matters because if price knowledge were solely driven by traits such as cognitive ability or shopping patterns, reducing search costs through better access to price information might have limited effect. Instead, our results indicate that knowledge is shaped by information access and that interventions aimed at improving the accessibility and visibility of price information—such as price disclosure regulations or comparison tools—may be effective in raising consumer knowledge and affecting market outcomes.

The prediction that informed consumers lead to lower prices and increased competition underpins a large empirical literature on the effects of price transparency regulations.

[Ater and Rigbi \(2023\)](#) study the effects of a price transparency regulation implemented

in the food retail industry in Israel in 2015. The regulation required supermarkets to submit daily prices to an online repository, which was then used by independent websites to provide consumers with free price-comparison services. The authors find that average prices declined by 4 to 5 percent following the price transparency regulation. The authors identify the decreasing cost and increasing popularity of large-scale price comparison surveys by media outlets following the regulation as a likely mechanism driving the price reduction. Focusing on medical imaging procedures, [Brown \(2019\)](#) studies the introduction of a state-wide price transparency website and finds that price information led to lower prices and that the effect is driven both by demand-side (patients shifting to lower-cost providers) and supply side (firms charge lower prices) responses. In addition, several studies have investigated the impact of price transparency measures in gasoline markets finding somewhat diverging results, [Rossi and Chintagunta \(2016\)](#); [Montag et al. \(2024\)](#); [Luco \(2019\)](#). While this line of literature typically focuses on market-level outcomes—such as reduced average prices or lower price dispersion—our study provides direct micro-level evidence for the link between price knowledge and prices.

We also contribute to the marketing literature on price knowledge and consumer behavior. Prior studies in this field have primarily measured price recall and examined its implications for pricing strategies and the marketing mix ([Winer, 1986](#); [Dickson and Sawyer, 1990](#); [Loy et al., 2020](#)), but have generally lacked data linking recall to actual purchase behavior. We address this gap by systematically combining survey-based price knowledge data with transaction histories from the same consumers. Our results show that much of the variation in price knowledge is explained by differences in shopping behavior, highlighting the value of behavioral data for understanding who is more or less informed.

2 Data

2.1 Data Sources

We use four primary sources of data. The first is a digital survey, distributed with the help of NorgesGruppen — Norway’s largest grocery umbrella chain — among 50 000 members of the customer loyalty program Trumf, of whom 2505 responded. After being asked about demographics, the respondents were asked to recall today’s shelf price, accounting for sales but ignoring individual discounts, of 18 out of 24 representative grocery products for two grocery chains belonging to the umbrella chain, the leading discount chain and the leading supermarket in Norway.³ The chosen products cover the major categories of groceries and should be familiar to the vast majority of Norwegian grocery shoppers. Finally, they were

³By asking for specific numbers we are placed in the part of the literature on price knowledge focusing on explicit, as opposed to implicit, price knowledge (see [Evanschitzky et al. \(2004\)](#)).

asked to state their degree of agreement on 10 different statements meant to elicit their subjective degree of price consciousness. The complete survey and more details can be found in Appendix D. Out of the 2505 respondents, 2285 made at least one price recall, resulting in 77 646 different price recall observations. Note that there is a unique recall for every individual $\times$ chain $\times$ product-combination, whereas the other survey questions only vary across individuals. The data is unbalanced both because of missing responses and each respondent only being asked about 18 out of the 24 products included in the survey.

The second data source is the loyalty-card based daily purchasing history in this grocery umbrella chain — spending and quantities — of each respondent from January 1st 2023 to June 12th 2024 for the products included in the survey in addition to their close substitutes, amounting to over 500 products. The dimensions of the daily sales data are: individual $\times$ chain $\times$ products $\times$ date, and is highly unbalanced since most people do not buy every product, every day, at both chains.

The third data source is weekly purchasing frequencies, spending, and the number of unique products purchased at both grocery chains, including products not included in the daily data. The weekly sales data has the following dimensions: Individual $\times$ chain $\times$ week.

The fourth data source consists of the most frequently paid (modal) price for each chain $\times$ product $\times$ municipality $\times$ date-combination for the products included in the daily sales data, which nearly always corresponds to the shelf price.

In order to relate the sales and price data to the survey data, we aggregate various variables based on the purchasing histories, such as spending, purchasing frequencies, and prices paid, over time and product categories before merging. When appropriate, we further aggregate the data to the individual level, for instance by computing the average error rates for every chain $\times$ product-combination that the individual recalled the price of, or by calculating the individuals’ overall discount utilization. Similarly, we calculate average price recall error rates for the different chains and products to see how the respondents’ price knowledge varies across them, and whether they tend to overestimate or underestimate the prices of certain chains or products.

2.2 Defining Some Key Variables

2.2.1 Price Knowledge

The model of sales model proposed by [Varian \(1980\)](#) assumes two distinct consumer types: perfectly informed consumers, who know the full distribution of prices and always select the lowest available price, and completely uninformed consumers, who choose at random. Formally, the model is a static price game with mixed strategies, though it can also be interpreted dynamically, where mixed strategies represent random sales occurring over time. A key prediction of the model is that informed consumers, by identifying and acting

on price variation, pay lower average prices than uninformed consumers.

The binary distinction between informed and uninformed consumers is clearly a simplification, even if empirically there exist mass points of certain types. Generally, being informed about prices is a matter of degree. For the theory to be practically useful, it should be interpreted as implying that greater price knowledge leads to lower average prices. To test whether price knowledge affects prices paid, we therefore construct a continuous measure of price knowledge based on consumers' ability to accurately recall current prices for a range of store-product combinations. This measure is then aggregated across store-product combinations to derive an overall price knowledge score for each consumer.

Our primary measure of price knowledge is based on the Percentage Absolute Deviation (PAD) of the surveyed price recall of individual i for product j at chain c relative to the true price in home municipality m at survey response date d :

$$\text{PAD}_{ijc} = \left| \frac{P_{ijc}^{\text{Recall}} - P_{jcmd}^{\text{True}}}{P_{jcmd}^{\text{True}}} \right| \quad (2.1)$$

The PAD measure has previously been used in the marketing literature on price knowledge (see Dickson and Sawyer (1990); Estelami (1998); Wakefield and Inman (1993) and Zeithaml (1984)). By surveying respondents outside of purchasing situations, as opposed to studies surveying consumers post-purchase, in-store (see Conover (1986); Dickson and Sawyer (1990); Le Boutillier et al. (1994); McGoldrick and Marks (1987) and Wakefield and Inman (1993)), we aim to capture the long-term price memory of the respondents (see Evanschitzky et al. (2004) for a discussion about short vs. long-term memory and survey timing). We define $PK_{ijc} = -PAD_{ijc}$ as our measure of price knowledge. PK_{ijc} is weakly negative and an increase in PK_{ijc} represents an increase in price knowledge towards zero, i.e. perfect information.

To get a measure of the individual's overall price knowledge we calculate the average percentage error that the individual made in the survey:

$$\text{PK}_i = \frac{1}{K_i} \sum_{k_i=1}^{K_i} \text{PK}_{ijc} \quad (2.2)$$

where K_i is the number of product and chain combinations that the individual recalled prices for in the survey.⁴

To complement the PK we can also indicate whether a recalled price is within some

⁴Equation 2.1 measures price knowledge without distinguishing between overestimation and underestimation. In the interest of measuring the direction of the respondents' price bias, we can calculate $\text{PD}_{ijc} = \frac{P_{ijc}^{\text{Recall}} - P_{jcd}^{\text{True}}}{P_{jcd}^{\text{True}}}$. The respondent's overall price bias is then given by $\text{PD}_i = \frac{1}{K_i} \sum_{k_i=1}^{K_i} \text{PD}_{ijc}$. Finally, averaging indicators of the sign of PD_{ijc} lets us calculate the respondent's share of over and under-estimations.

percentage error bound e : $W_{ijc}^e = \mathbf{1}\{-PK_{ijc} \leq e\}$, where $e \in [0, 1]$. This measure has previously been used in the marketing literature on price knowledge (e.g. [Goldman \(1977\)](#) and [McGoldrick and Marks \(1987\)](#)), and has the benefit of being robust to extreme outliers in recalled prices. We will see that the cases where a consumer is able to recall a product price perfectly ($e = 0$) are vanishingly rare, but using $e = 0.2$ gives us a measure that is strongly correlated with PK. The share of “correct” recalls for various degrees of error leniency can be calculated as follows $W_i^e = \frac{1}{K_i} \sum_{k_i=1}^{K_i} W_{ijc}^e$.

2.2.2 Price Consciousness

The survey also includes questions to elicit the respondents’ price consciousness, as an underlying trait, by asking them to state their level of agreement on a 5-point Likert scale. The exact statements can be found at the end of the survey in Appendix D and the distributions of the answers for each question are shown in Figure C.5.⁵ The first 3 questions deal with the degree to which the respondents incorporate prices in their purchasing decisions, like for example how important price is for their choice of grocery store or products. Questions 4 to 7 pertain to various activities that involve monitoring and comparing grocery prices, for example checking shelf prices or reading grocery chain flyers.⁶ In combination, questions 1 to 7 examine the respondents’ price consciousness, both in terms of information gathering and decision-making.

In order to capture different aspects of price consciousness, we calculate indices by averaging over the Likert scale scores. In the following, we label the average over questions 1 to 3 as “Price Sensitivity”, the average over questions 4 to 7 as “Price Learning Effort”, and the average over questions 1 to 7 as “Price Consciousness”. To allow for certain Likert scale questions being more or less important in explaining the underlying trait, we calculate weighted averages based on the estimated factor loadings from a simple factor model: $Likert\ Response_i = \tau_i + \lambda_i Factor + \epsilon_i$. Here, i indicates a Likert Scale item, τ_i ’s is an intercept, λ_i is the factor loading, ϵ_i is the residual, and $Factor$ is one of the underlying factors: price consciousness, price sensitivity, or price learning effort. The factor loadings of each Likert scale question for the underlying factors, estimated using diagonally weighted least squares, are presented in Table C.1 in the Appendix. The weight attributed to each Likert item is found by dividing its factor loading by the sum of all the factor loadings.⁷

2.2.3 Discount Utilization

The products in [Varian \(1980\)](#) are assumed to be homogeneous, making it straightforward to compare prices paid by different consumer groups. However, in grocery retailing,

⁵[Loy et al. \(2020\)](#) uses similar, but fewer, statements and a 7-point Scale.

⁶The last 3 questions are about how confident the respondents are in their own price knowledge.

⁷We also calculate simple averages over the Likert items and find that the results are largely the same.

products and stores are differentiated, and consumers may pay lower prices not only due to price knowledge but also due to preferences for lower-priced stores or products. Our detailed data on purchasing history allows us to calculate an individual-level price measure that strips out variation stemming from differences in the overall price level of the store and products.

To test the prediction that more informed consumers pay lower prices on average and to measure the gains associated with knowledge of grocery prices, we calculate Discount Utilization (DU), the percentage difference between the average daily (modal) price in the consumer's home municipality m ($\bar{P}_{jcme}^{\text{Modal}}$) over some event window e and the price a consumer i pays on a given day t for product j at chain c (P_{ijct}^{Paid}):

$$\text{DU}(E)_{ijct} = \frac{\bar{P}_{jcme}^{\text{Modal}} - P_{ijct}^{\text{Paid}}}{\bar{P}_{jcme}^{\text{Modal}}} \quad (2.3)$$

where E marks the length of the event windows. This measure indicates whether a given purchase is a bargain compared to buying the product randomly throughout the event window with equal probability placed on each day.

Of course, the scope for paying lower prices depend on the event window. Since traditional sales apply to everyone that buys a product on the same day, the only way to find intra-day bargains is to use personal coupons or some quantity or expiration date discount. Conversely, over longer periods like 1-3 months, you can pay less than the average shelf price by timing purchases with sales. We have 18 months of purchasing history, but Equation 2.3 might pick up unbalanced spending over time combined with general inflation, so we focus on a 3-month event window ($E = Q$). This limits the effects of inflation and leaves ample room for timing sales.⁸

In order to calculate the overall tendency of a consumer to buy specific products at lower prices, we average 2.3 over time:

$$\text{DU}(E)_{ijc} = \frac{1}{T_{ijc}} \sum_{t=1}^{T_{ijc}} \text{DU}(E)_{ijct} \quad (2.4)$$

where T_{ijc} is the number of times that an individual i is observed purchasing product j at chain c in the data. This measures how much less an individual tends to pay for specific products than someone buying the same product, but timing their purchases uniformly throughout the event windows e that the individual's purchases fall into.

To get an overall measure of a consumer's ability to identify and act on bargains, we further average Equation 2.4 across product and chain combinations:

⁸The results are similar using 1, 3, or 18 months ($E \in \{M, Q, FP\}$), but are strongly muted using daily discount utilization ($E = D$), as expected.

$$DU(E)_i^{\text{Simple}} = \frac{1}{N_i} \sum_{n_i=1}^{N_i} DU(E)_{ijc} \quad (2.5)$$

where N_i is the number of product and chain combinations that we observe individual i purchasing in the data. This gives us a simple measure that quantifies an individual's tendency to exploit price variation over time for all the different products that they buy (and obviously not only the products for which they later have been asked to produce price calls).

Note that we do not average over every transaction, which would over-weight frequently purchased products. The reason for this is that you could identify a sale once and then act on it several times on consecutive days. We are primarily interested in the lower prices one can achieve because of more price knowledge. However, we also calculate a sales-weighted version of Equation 2.5,⁹ which allows us to calculate the percentage savings that the consumer makes by buying its current basket of groceries on sales relative to buying the same basket uniformly over time. Since sales weights also are sensitive to purchasing frequency, we focus on Equation 2.5 when we test and discuss the theoretical predictions of models such as [Varian \(1980\)](#)'s model of sales pertaining to price knowledge and average prices paid, and we focus on Equation 2.5 in the interest of calculating the gains associated with price knowledge.

2.2.4 Price Variation and Scope for Discount Utilization

Groceries constitute many different products with varying levels and dispersion in price, and the scope for buying products on sale critically depends on the sales activity and overall price variation of the products you buy. To capture and control for this feature of grocery prices, we calculate the coefficient of variation for each product $j \times$ chain $c \times$ municipality m -combination over the sample period:

$$CV_{jcm} = \frac{\sigma_{jcm}}{\mu_{jcm}} \quad (2.7)$$

where σ_{jcm} is the empirical standard deviation of price and μ_{jcm} is the product's mean price over the sample period. There is certainly more room for discount utilization in products with high coefficients of variation but the relation between knowledge of a product's price and its variation is less clear. Theoretically (see [Stigler \(1961\)](#)), we know

⁹Formally, we compute

$$DU(E)_i^{\text{Sales-Weighted}} = \sum_{n_i=1}^{N_i} \frac{S_{ijc}}{S_i} DU(E)_{ijc} \quad (2.6)$$

where S_{ijc} is the consumer's spending on each chain $\times$ product-combination and S_i is the consumer's total spending, over the sample period.

that the gains to knowing prices increases with price dispersion, but at the same time, it is likely more difficult to keep track of product prices that vary a lot.

At the individual level, it is useful to consider the overall price variation in the consumer's shopping basket. Consumers that primarily buy products with little price variation over time are more limited in their opportunity to find bargains, and vice versa. We calculate the average price variation of the products that individual i buys in the following way:

$$CV_{i[\text{Simple}]} = \frac{1}{N_i} \sum_{n_i=1}^{N_i} CV_{jcm} \quad (2.8)$$

where N_i is the number of product and chain combinations that we observe individual i purchasing in the data. Similarly, the price variation of the consumer's average shopping basket is given by:

$$CV_{i[\text{Sales-Weighted}]} = \sum_{n_i=1}^{N_i} \frac{S_{ijc}}{S_i} CV_{jcm} \quad (2.9)$$

where S_{ijc} is the consumer's spending on each chain $\times$ product-combination and S_i is the consumer's total spending, over the sample period. We use Equation 2.8 when we analyze simple discount utilization and Equation 2.9 when we analyze sales-weighted discount utilization.

2.3 A First Look at the Data

For comprehensive descriptive statistics and details on data cleaning, we refer the reader to Appendix A. Here, we highlight and discuss the key variables central to our analysis.

First, to assess the external validity of our results, it is useful to consider the composition of the survey sample. While the sample is broadly representative in terms of demographics, there are some noteworthy deviations.¹⁰ Compared to the general population in 2024, respondents are more likely to have higher education (71% vs. 38%), to be over 40 years of age (77% vs. 51%), and to be female (62% vs. 50%).¹¹

The average value of the price consciousness index derived from the survey is 3.81. The sub-indices for price sensitivity and learning effort have averages of 4.12 and 3.52,

¹⁰The survey is collected from a random sample of 50 000 Trumf members. Trumf has nearly 3 million members of an adult population of 4.5 million (80% out of total of 5.6 million are over 18 years old, <https://www.ssb.no/statbank/table/10211/tableViewLayout1/>), being the largest Norwegian loyalty program, and the grocery chain considers them to be representative for all their customers. In fact, earlier the chain tracked both individual bank card activity and Trumf members' activity, but now they have stopped doing the former, since they do not find any noteworthy differences between these two groups now as Trumf membership has become so widespread. This implies e.g., that there is no reason to believe that the Trumf customers are more price sensitive than the average population.

¹¹Population benchmarks from [Statistics Norway \(2024a\)](#) (age and gender) and [Statistics Norway \(2024b\)](#) (education).

respectively. This suggests that most respondents *agree* or *strongly agree* with statements related to price sensitivity, while views on activities that generate price knowledge are more mixed. The two sub-indices are moderately correlated, with a coefficient of 0.59.

We use several indicators of price consciousness, some based on survey responses and others derived from transaction data. If these measures capture a common underlying construct, they should be positively correlated. In particular, we examine whether the survey-based index aligns with behavioral indicators. Reassuringly, the correlations are generally positive (see Figure A.1). Discount utilization and the share of spending on low-cost private labels—both from transaction data— show the strongest associations, with coefficients of 0.32 and 0.33, respectively.

The empirical distribution of individual price knowledge, captured by PK, is shown in the left panel of Figure 2.1. The mean PK is -0.27 , indicating that, on average, respondents either overestimate or underestimate the true price by 27%. The interquartile range spans from -0.32 to -0.22 . Seventeen percent of recalled prices are within 5% of the true price, and 50% are within 20%.¹² The distribution shows no distinct mass points that would suggest discrete consumer information types. Moreover, no respondent recalls prices perfectly—the minimum PK is -0.04 . We also note that among all recalled prices included in the analysis, 31% were provided by respondents who had purchased the product at the relevant chain during the sample period. For these cases, the average time since purchase is 4.7 months, and the median is 2.7 months.

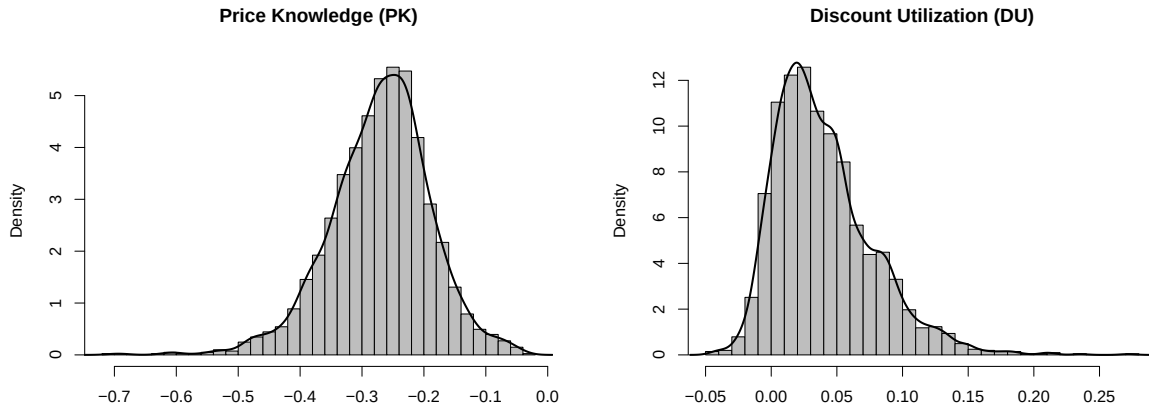


Figure 2.1: Histograms and density estimates of price knowledge and discount utilization. Histogram bin sizes are chosen using Scott’s rule and density bandwidths are chosen using Sheather-Jones’ plug-in estimator.

The distribution of discount utilization, computed as the simple average across products, is shown in the right panel of Figure 2.1. A value of zero implies behavior consistent with the uninformed consumers in Varian’s model—those who time purchases randomly

¹²Our findings are broadly in line with those of Loy et al. (2020), who report an average PK of 35% and a share of recalled prices within 5% of the true price of around 14% in a survey conducted in Northern Germany.

with respect to sales. Most consumers, however, exhibit positive discount utilization: the interquartile range is 0.01 to 0.06, and the mean is 0.04, indicating that consumers tend to concentrate purchases on days when prices are below average. The mean of the simple average metric exceeds the sales-weighted version (2%), suggesting that consumers do not always obtain the largest discounts on the products they spend the most on. One possible explanation is that more expensive products are discounted less frequently or less deeply.

Finally, we observe a positive raw correlation of 0.31 between PK and discount utilization. This relationship is the focus of the analysis in the following section.

3 Price Knowledge and Discount Utilization

The central prediction of Varian’s model of sales is that periodic and unpredictable sales is a price equilibrium when firms face consumers with different levels of information about prices. This happens because firms both want to exploit uninformed consumers that buy irrespective of price, and to attract informed customers that only choose the cheapest option. We posit that the key feature of Varian’s model driving this outcome is that some consumers are better informed about prices than others, making them more likely to select lower-priced products given a distribution of prices, and thus pay lower prices on average. If we can confirm this monotone relation between price knowledge and average prices paid, i.e. discount utilization, we lend credence to the predictions made by Varian’s model and the subsequent work that build on it.

3.1 Binned Scatterplots

To get a nonparametric estimate of the expected value of discount utilization conditional on price knowledge, we calculate a binned scatterplot using the implementation of [Cattaneo et al. \(2019\)](#), presented in Figure 3.1. Each point is the average discount utilization among the respondents with price knowledge within the corresponding bin. The bins are chosen to minimize the integrated mean squared error, and the 95 % confidence bands are calculated using piecewise polynomials.

Figure 3.1 gives a clear indication that respondents with more price knowledge also purchases more on sale and pay lower prices on average, consistent with [Varian \(1980\)](#)’s model of sales, since a flat line cannot fit within the confidence bands, whereas an increasing line can. Although there is a slight S-shape to the estimated conditional expectation function, most pronounced at the edges, we cannot reject a linear relation.

3.2 Linear Regression Analysis

In order to parametrize the relation between discount utilization and price knowledge, and control for confounding variables, we consider regression models of the following form:

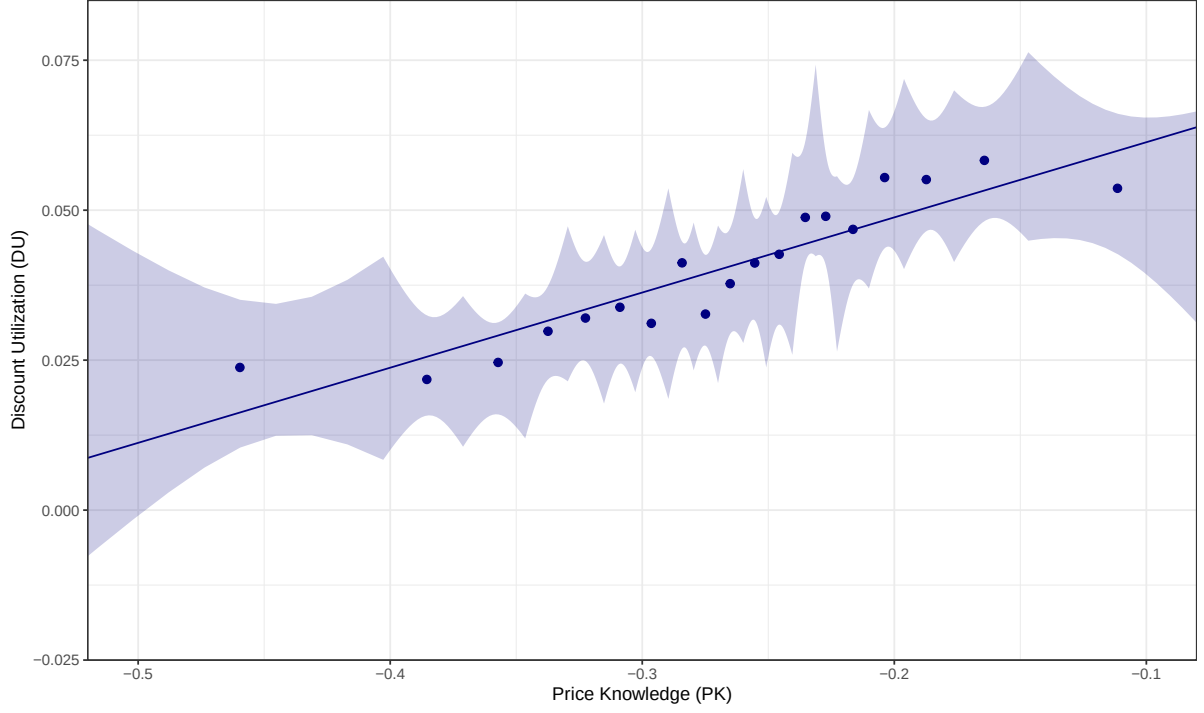


Figure 3.1: Estimate of the expected value of discount utilization conditional on price knowledge using a binned scatterplot with a linear fit. The number of bins are chosen to minimize the integrated mean squared error. Confidence bands at 95 % are calculated using piecewise polynomials.

$$DU(Q)_i^{\text{Simple}} = \alpha + \delta \cdot PK_i + \beta' \mathbf{X}_i + \epsilon_i \quad (3.1)$$

where $\mathbf{X}_i$ is a vector of control variables, ϵ_i is the regression error, and $\{\alpha, \delta, \beta\}$ are the parameters to be estimated.

The OLS estimates for equation 3.1 with different control variables are presented in Table 3.1.

Table 3.1: Discount utilization regressions

	Discount Utilization		
	(1)	(2)	(3)
Price Knowledge	0.125*** (0.011)	0.121*** (0.010)	0.087*** (0.009)
Male		0.000 (0.002)	0.001 (0.002)
Single Person Household		0.004 (0.003)	0.004 (0.003)
Has Children		0.004* (0.002)	0.005** (0.002)
Full-Time Employment		-0.005* (0.002)	-0.003 (0.002)
Higher Education		0.001 (0.002)	0.002 (0.002)
Rural		-0.006*** (0.002)	-0.005** (0.002)
Household Income > 950' NOK		-0.002 (0.002)	-0.001 (0.002)
Age: 40-59 Years		0.007*** (0.002)	0.007*** (0.002)
Age: Above 60 Years		0.008** (0.003)	0.009*** (0.002)
Price Sensitivity		0.007*** (0.001)	0.003* (0.001)
Price Learning Effort			0.008*** (0.001)
# Frequented Grocery Chains			0.000 (0.001)
Mainly Uses Discount Stores			-0.003 (0.002)
Average Weekly Store Visits			0.003*** (0.001)
Main Grocery Shopper			0.002 (0.002)
Average Weekly Variety (per 10 products)			-0.003*** (0.001)
Three Customer Program Memberships			0.005*** (0.001)
Basket Coefficient of Variation			1.216*** (0.074)
Constant	0.074*** (0.003)	0.041*** (0.006)	-0.116*** (0.010)
Observations	2,028	1,935	1,928
Dependent variable mean	0.04	0.04	0.04
R ²	0.07	0.12	0.35

Notes: This table reports results from estimation of Equation ???. The dependent variable, individual-level discount utilization, is the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product $\times$ chain combinations (simple weights). Price knowledge is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. The average weekly variety is calculated as the average number of unique products purchased per week over the sample period. Price sensitivity and learning effort are calculated as the factor-weighted averages of Likert-scale survey items 1–3 and 4–7, respectively. Basket CV is calculated as the coefficient of variation of shelf prices over the sample period, averaged across all products in the consumer's purchase history, using simple weights. Column 1 reports the simple regression on price knowledge (PK), Column 2 adds controls for demographics and price sensitivity, and Column 3 adds controls for shopping habits and price learning effort. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

A simple regression, presented in Column 1, shows that a 10 percentage point increase in price knowledge is associated with a 1.3 percentage point increase in discount utilization, a change in price knowledge that is well within the variation in the data; the average PK in the sample is -27 %, the standard deviation is 8%, and 50 % of respondents have a PK between -22 and -32 %. Someone with very good price knowledge (e.g. $PK_i = -10$ %) pays on average 4 % less for their groceries than someone with very poor price knowledge (e.g., $PK_i = -40$ %). The regression constant is also of theoretical interest. In fact, assuming that a linear extrapolation to the point of $PK_i = 0$ is reasonable, it gives us the expected discount utilization of a perfectly informed consumer. At 7.4%, the discount utilization of a perfectly informed consumer is higher than the average of 4%, and substantially higher than the reference-level of discount utilization associated with the “uninformed” consumers in Varian’s model at 0%.

To examine robustness with respect to potential confounders, Columns 2-4 in Table 3.1 introduce variables intended to control for different proclivities for shopping on sales that may correlate with price knowledge. Introducing demographic variables, shown in Column 2, does not meaningfully alter the coefficient on PK, which is consistent with most demographic variables not being strongly correlated with either discount utilization or price knowledge (see Figure A.2). In terms of explaining discount utilization, the R^2 only increases modestly, but the coefficients generally have the expected signs: People with children, low incomes and not full employment pay lower prices, as do older people and those living in urban areas.

Column 3 includes variables related to the consumers’ shopping habits and preferences for price, which more than triples the R^2 from the demographics-only specification. The *ceteris paribus* association between price knowledge and sales utilization is now somewhat muted, largely due to the indicator for loyalty memberships and especially the self-reported price consciousness index, which are more correlated with both discount utilization and price knowledge than other variables. The price consciousness index ranges between 1 and 5, and a unit increase ($\sim$ the length of the inter-quartile range) is associated with 1.1 % lower prices paid, and being a member of all the main customer loyalty programs is associated with paying 0.5% lower prices. By including markers of price consciousness, we control for the respondents’ general taste for buying on sales and discounts, which plausibly affect both price knowledge acquisition and discount utilization. Crucially, the results suggests that price knowledge still is an independent prerequisite for identifying, and then acting on, good deals.

Although it has minimal bearing on the coefficient on PK, a large portion of the gain in R^2 is due to the inclusion of the average coefficient of variation of the products purchased by the respondent, with a standard deviation increase of 0.01 in basket price variation being associated with 1.3 % lower prices paid. After all, the possibility of finding a bargain hinges on variation in prices over time.

The coefficients on the remaining variables related to shopping habits are smaller in magnitude, but buying many different products per week is associated with less discount utilization, and shopping frequently is associated with more discount utilization. Going to the store more frequently naturally increases your opportunity to act on sales. Regarding weekly variety, it could be that shoppers with a preference for variety also are less price conscious, but also that regularly buying more products necessarily means that you also have to buy products that are not on sale at a given time.¹³

Column 4 combines controls for demographics and shopping habits. Most coefficients from Columns 2 and 3 are largely unaffected, but the effects of household income and full-time employment are now muted, presumably because their relation to discount utilization are captured by markers of price consciousness. In terms of both average contribution to R^2 and standardized coefficients (see Table C.2 in the Appendix), the variables in Column 4 that are the most important in explaining discount utilization are the basket coefficient of variation, price consciousness, price knowledge, and weekly store visits, in that order.

The primary takeaway from Table 3.1 is that even after controlling for variables that signal an interest in, and an ability to act on, temporary price discounts, the positive association between price knowledge and discount utilization remains. This supports a foundational assumption of Varian (1980)’s model of sales and by extension the vast amount of both theoretical and empirical work that builds on the model.

3.3 The Gains Associated with Price Knowledge

In Varian (1980)’s model, and similar search-theoretic models of industrial organization, increasing price knowledge in the population of consumers increases consumer welfare in two ways. First, the direct effect at the level of the consumer of becoming informed about prices is that the consumer will be able to enjoy paying lower prices facing a given distribution of prices. Secondly, more informed consumers, which are sensitive to price differences between options, leads to tougher competition and subsequently lower prices in equilibrium. We cannot speak to the equilibrium effects of increasing price knowledge on prices with the data and methodology that we employ in this paper, but we can perform back-of-the-envelope calculations to quantify the direct gains associated with increased price knowledge, taking current prices and quantities as fixed.

To do this we rely on the sales-weighted version of discount utilization, which allows us to quantify the savings a consumer is expected to make by buying its current basket of goods at more opportune times. Specifically, we estimate again a model like the one described in Equation 3.1 but where we use $DU(Q)_i^{\text{Sales-Weighted}}$ as the left-hand side

¹³If we instead count the number of different products purchased over the whole sample period, the sign flips. You might buy many different products over time because you jump on discounts when you see them.

variable.

The full results are shown in Table C.4 in Appendix C.¹⁴ The results are qualitatively similar to the simple average version, except that mainly shopping at discount stores now has a detectable and negative effect. However, the coefficients are muted in magnitude across the board since the opportunity space for paying lower prices than the average for a given product is determined by its variation in price.

If we consider a nuclear family of a mother (31-60 years), father (31-60 years), daughter (10-13 years), and son (2-5 years), their reference budget for groceries in 2024 according to SIFO (2024) amounts to 173 640 NOK per year. With fixed quantities, a 30 percentage point decrease in PK, a substantial but feasible change in light of its empirical distribution (see Figure 2.1), is associated with yearly household savings of 3629 NOK (2.1%).

It is also instructive to consider the savings that a perfectly informed consumers is expected to make relative to a consumer that makes its purchases randomly throughout the quarter akin to the uninformed consumers in Varian (1980). Assuming that a linear extrapolation of the relation between price knowledge and sales utilization to the point of $PK = 0$ is reasonable, the sales utilization of a perfectly informed consumer is given by the regression constant, which is 4% in the simple regression. This suggests that a perfectly informed consumer shopping on behalf of the nuclear family pays 4 % less for its groceries than someone that does not time their purchases with temporary price discounts, amounting to yearly household savings of 6925 NOK.

The potential household-level savings related to improvements in price knowledge are quantitatively noteworthy but not enormous. However, from a policy perspective, improvements in price knowledge adds up across households and over time. Low-cost initiatives that persistently improve the price knowledge of the broader population could lead to large welfare gains. Furthermore, the search-theoretic prediction of an increase in the share of informed consumers, the foundations of which we argue this article provides empirical support for, is that equilibrium prices will fall. The direct savings from better price knowledge that we calculate here should therefore be seen as a lower bound on the total savings to consumers.¹⁵

¹⁴See also Figure C.6 and for a binned scatterplot using sales-weighted discount utilization and price knowledge. The overall pattern is the same as when using simple averages across purchased products, but more compressed. As discussed earlier, this is not surprising since the products that the respondents spend more on vary less in price over time, reducing the scope for bargain-hunting. From Figure C.7, we can see that this leaves us with a compressed version of the distribution of discount utilization based on simple averages.

¹⁵Most people in our sample buy from one of the leading discount chains in Norway, arguably the one with the least price variation over time among the grocery chains. The gains to price knowledge can only be expected to be larger for the general population, some of which buy groceries from chains with more sales activity. This further strengthens the conclusion that our calculations are lower bounds on the gains associated with price knowledge.

3.4 Heterogeneity

As discussed above, discount utilization is increasing on average in the price variation of the products that the respondents buy, which is evident from the top right panel of Figure C.3 in Appendix C. It is also the case that there is more variation in discount utilization among the consumers that buy products with a lot of price variation, so if price knowledge affects discount utilization and savings on groceries, it has more potential to do so for consumers that tend to buy products with a lot of price variation.

Table 3.2 contains estimates of separate regressions of price knowledge on discount utilization for respondents above and below the median in terms of basket price variation. The interaction term shows that the association between price knowledge and discount utilization is stronger for those with above median price variation in their average shopping baskets. The yearly savings associated with a 30 percentage point increase in PK for the below median consumers is now only 2396 NOK (1.4%), but it is 4428 NOK (2.6%) for the above median consumers. Similarly, the savings relative to a consumer that disregards sales altogether, is 4167 NOK (2.4%) below the median and 9203 NOK (5.3%) above the median.

The empirical finding that the gains from price knowledge increase with price variation aligns with the fundamental prediction first proposed by Stigler (1961) that the gains from search are increasing in the product's price dispersion (interpreted here as variation over time), in the sense that in Stigler's model, search entails increasing price knowledge by learning the price of an additional product.¹⁶

¹⁶The expected minimum price paid for a product after sampling n sellers when prices are uniformly distributed on $p \in [0, b]$ is equal to $E[p^{\text{Min}}] = \frac{b}{n+1}$. The gain from more search n is then $\frac{\partial |E[p^{\text{Min}}]|}{\partial n} = \frac{b}{(n+1)^2}$, which is increasing in price dispersion b : $\frac{\partial^2 |E[p^{\text{Min}}]|}{\partial n \partial b} = \frac{1}{(n+1)^2} > 0$.

Table 3.2: Discount utilization regressions: basket CV interactions

	DU ^{Simple}		DU ^{Sales-Weighted}	
	(1)	(2)	(3)	(4)
Price Knowledge (PK)	0.125*** (0.011)	0.087*** (0.010)	0.070*** (0.010)	0.046*** (0.005)
High Basket CV (Simple)		0.041*** (0.006)		
Price Knowledge (PK) $\times$ High Basket CV (Simple)		0.053** (0.020)		
High Basket CV (Sales-Weighted)				0.029*** (0.005)
Price Knowledge (PK) $\times$ High Basket CV (Sales-Weighted)				0.039* (0.018)
Constant	0.074*** (0.003)	0.050*** (0.003)	0.040*** (0.003)	0.024*** (0.002)
Observations	2,028	2,028	2,028	2,028
Dependent variable mean	0.04	0.04	0.02	0.02
R ²	0.07	0.20	0.04	0.16

Notes: This table reports results from estimation of discount utilization regressions using an interaction between price knowledge (PK) and an indicator for having above-median basket price variation. Discount utilization is calculated as the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product $\times$ chain combinations, with the latter using simple and sales weights. Price knowledge (PK) is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. Basket CV is calculated as the coefficient of variation of shelf prices over the sample period, averaged across all products in the consumer's purchase history, using simple and sales weights. Column 1 includes the simple regression on price knowledge (PK), Column 2 adds an interaction with Basket CV, and Columns 3–4 repeats the estimation using sales-weighted discount utilization. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

It is also useful to see if the gains associated with price knowledge differ based on price consciousness. It could be that consumers that really care about prices are more likely to utilize price knowledge to pay lower prices. Table 3.4 shows regression estimates for price knowledge, price consciousness, and their interaction using simple and sales weights.¹⁷ We find that the effect of price knowledge is increasing in price consciousness, suggesting that those that care a lot about price are more likely to act on information about prices. From the summary statistics on price consciousness in Table A.1, we calculate the marginal effect of price knowledge for different levels of price consciousness:¹⁸

Table 3.3: Marginal Effects of Price Knowledge Conditional on Price Consciousness

	Min	1st Quantile	Median	Mean	3rd Quantile	Max
Price Consciousness	1	3.36	3.91	3.81	4.38	5
Effect of Price Knowledge (Simple)	0.025 (0.032)	0.092*** (0.012)	0.107*** (0.012)	0.105*** (0.012)	0.121*** (0.015)	0.138*** (0.020)
Effect of Price Knowledge (Sales-Weighted)	0.028 (0.025)	0.054*** (0.010)	0.060*** (0.011)	0.059*** (0.011)	0.065*** (0.013)	0.072*** (0.017)

Notes: This table reports average and quintile marginal effects of price knowledge on discount utilization. The standard errors shown in parentheses are robust to heteroskedasticity (HC1) and calculated using the delta method. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The inter-quartile difference in the marginal effect of price knowledge on discount utilization conditional on price consciousness is $\sim 3\%$. This finding has potential implications for the distributional effects of policies aimed at improving price knowledge. In our sample, the most price conscious are often younger women with children and lower incomes (see Figure A.2 in Appendix C). To the extent that disadvantaged groups are more price conscious, population-wide gains in price knowledge may disproportionately benefit them, making such policies progressive.

¹⁷The estimating equation is $DU(Q)_i = \alpha + \delta \cdot PK_i + \beta_1 \cdot PC_i + \beta_2 \cdot PK_i \cdot PC_i + \epsilon_i$

¹⁸The marginal effect is $\frac{\partial DU(Q)_i}{\partial PK_i} = \delta + \beta_2 \cdot PC_i$, with standard error $SE = \sqrt{\text{Var}(\delta) + PC_i^2 \cdot \text{Var}(\beta_2) + 2PC_i \cdot \text{Cov}(\delta, \beta_2)}$, and t -statistic $t = \frac{\delta + \beta_2 \cdot PC_i}{SE}$. The only non-significant marginal effect is for the minimum value of the price consciousness index. However, only *one* respondent in our sample report a price consciousness coefficient of one, less than 40 respondents, ($<0.02\%$) have reported a price consciousness coefficient between 1 and 2.

Table 3.4: Discount utilization regressions: price consciousness interactions

	DU ^{Simple}		DU ^{Sales-Weighted}	
	(1)	(2)	(3)	(4)
Price Knowledge (PK)	0.125*** (0.011)	-0.003 (0.044)	0.070*** (0.010)	0.017 (0.034)
Price Consciousness		0.021*** (0.004)		0.009*** (0.003)
Price Knowledge (PK) $\times$ Price Consciousness		0.028* (0.012)		0.011 (0.009)
Constant	0.074*** (0.003)	-0.011 (0.014)	0.040*** (0.003)	0.002 (0.011)
Observations	2,028	1,961	2,028	1,961
Dependent variable mean	0.04	0.04	0.02	0.02
R ²	0.07	0.14	0.04	0.07

Notes: This table reports results from estimation of discount utilization regressions using an interaction between price knowledge (PK) and price consciousness. Discount utilization is calculated as the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product $\times$ chain combinations, with the latter using simple and sales weights. Price knowledge (PK) is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. Price consciousness is calculated as the factor-weighted average of seven Likert-scale survey items regarding price sensitivity and learning effort. Column 1 includes the simple regression on price knowledge (PK), Column 2 adds an interaction with price consciousness, and Columns 3–4 repeats the estimation using sales-weighted discount utilization. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

4 Drivers of Price Knowledge

Having discussed and analyzed some of the gains associated with price knowledge, we now move on to the issue of understanding the drivers of price knowledge itself. The level of consumer price knowledge of groceries has been documented in many countries, including Great Britain (Gabor and Granger (1961); McGoldrick and Marks (1987)), Israel (Goldman (1977)), France (Vanhuele and Drèze (2002)), Finland (Kujala and Johnson (1993); Aalto-Setälä and Rajas (2003)), Germany (Diller (1988); Evanschitzky et al. (2004); Loy et al. (2020)), and the USA (Conover (1986); Dickson and Sawyer (1990); Manning et al. (2003)). We extend this literature by measuring consumer knowledge of grocery prices in Norway. Furthermore, previous studies in marketing (e.g. Loy et al. (2020) and ?) have focused on how price recall relates to measures that can be obtained from consumer surveys, like demographics, recited shopping habits and subjective price consciousness, or that are easily observed in stores by researchers, like current product prices and their dispersion. We add to those studies by relating price knowledge to variables that are derived from transaction data. This includes historic purchasing frequencies and spending on specific products, the number of store and chain visits, and other features of the respondents' shopping baskets, like product variety. Transaction data also allows us to measure variables like a product's historical price variation and number of close substitutes with great accuracy and completeness that is difficult to replicate by manual data collection.

We perform regressions, similar to Loy et al. (2020), of the following form:

$$\text{PK}_{ijc} = \theta + \gamma' \mathbf{X}_{ijc} + \epsilon_{ijc} \quad (4.1)$$

where PK_{ijc} is the recall-level error rate defined in Equation 2.1, $\mathbf{X}_{ijc}$ are explanatory variables, ϵ_i is the regression error, and $\{\theta, \gamma\}$ are parameters to be estimated. Note that not all variables in $\mathbf{X}_{ijc}$ vary across all of the dimensions in the data. For example, variables like price consciousness, shopping frequency, or gender only vary across individuals, and features of the products and their categories like price variation only vary across products $\times$ chain $\times$ home municipality combinations, whereas sales and purchasing frequencies are unique to each recalled price, i.e. each individual $\times$ product $\times$ chain combination.

Results from estimating Equation 4.1 with OLS are shown in Table 4.1.

Table 4.1: Price knowledge regressions

	Price Knowledge (PK)		
	(1)	(2)	(3)
Male	-0.015*** (0.004)	-0.015*** (0.004)	-0.015*** (0.004)
Single Person Household	-0.002 (0.007)	-0.002 (0.007)	-0.001 (0.007)
Has Children	0.002 (0.004)	0.003 (0.004)	-0.001 (0.004)
Full-Time Employment	-0.005 (0.004)	-0.005 (0.005)	-0.004 (0.004)
Higher Education	-0.013** (0.004)	-0.013** (0.004)	-0.012** (0.004)
Rural	0.000 (0.004)	0.000 (0.004)	-0.002 (0.004)
Household Income > 950' NOK	0.010* (0.004)	0.010* (0.004)	0.008* (0.004)
Age: 40-59 Years	0.008 (0.005)	0.008 (0.005)	0.006 (0.004)
Age: Above 60 Years	0.003 (0.006)	0.004 (0.006)	0.002 (0.006)
# Frequented Grocery Chains	0.002 (0.001)	0.002 (0.001)	0.002 (0.001)
Mainly Uses Discount Stores	0.004 (0.006)	0.006 (0.006)	0.004 (0.006)
Main Grocery Shopper	0.014** (0.004)	0.013** (0.004)	0.014** (0.004)
Supermarket Chain	-0.017*** (0.001)	-0.017*** (0.001)	0.033*** (0.002)
Private Label	0.044*** (0.004)	0.044*** (0.004)	0.018*** (0.004)
3 Customer Loyalty Program Memberships	0.011** (0.004)	0.011** (0.004)	0.010** (0.004)
Price Consciousness	0.019*** (0.003)		
Price Sensitivity		0.001 (0.003)	0.000 (0.003)
Price Learning Effort		0.016*** (0.002)	0.015*** (0.002)
Average Weekly Chain Visits			0.006*** (0.001)
Average Weekly Chain Variety (per 10 products)			-0.010*** (0.002)
Category Size (per 10 products)			-0.029*** (0.001)
Price Coefficient of Variation			-0.120*** (0.019)
Purchased			0.068*** (0.003)
Purchased $\times$ <i>Sales</i> in 1000 NOK			0.043*** (0.003)
Purchased $\times$ <i>Months Since Last Purchase</i>			-0.001** (0.000)
Constant	-0.354*** (0.013)	-0.344*** (0.013)	-0.308*** (0.014)
Observations	69,408	69,408	69,408
Dependent variable mean	-0.27	-0.27	-0.27
R ²	0.01	0.01	0.08

Notes: This table reports results from estimation of Equation ???. Price knowledge (PK) is calculated as the negative percentage absolute deviation between the respondents' recalled price and the true shelf price. Price consciousness is calculated as the factor-weighted average of seven Likert-scale survey items regarding price sensitivity and learning effort, while price sensitivity and learning effort are calculated similarly using Likert items 1–3 and 4–7, respectively. The average weekly chain variety is calculated as the average number of unique products purchased at each chain per week over the sample period. Category size is the chain-specific number of close substitutes for the product as defined by the grocery chain. The price coefficient of variation is calculated for each product $\times$ chain combination over the sample period. Purchased is an indicator of whether the respondent bought the product at a given chain during the sample period, and sales is the consumer's total product-chain spending. Column 1 includes individual characteristics and indicators for private labels and supermarket chain, Column 2 splits price consciousness into sensitivity and learning effort, and Column 3 adds variables derived from transaction data. The standard errors shown in parentheses are clustered by individual. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The standard errors are clustered at the individual-level, leaving us with 2028 clusters, to allow for persistency in how individuals recall the prices of different products.¹⁹ Column 1 only includes variables that are derived from the survey, Column 2 splits price consciousness into its two sub-factors price sensitivity and price learning effort, Column 3 adds variables that are derived from the transaction data.

We base the following discussion on the results in Column 3, which represents our preferred specification.

Demographics We see that men have poorer price knowledge than women. This replicates the findings of [Loy et al. \(2020\)](#) and [Estelami et al. \(2001\)](#), as opposed to [Estelami \(1998\)](#); [Jensen and Grunert \(2014\)](#); [Kenning et al. \(2011\)](#) and [Wakefield and Inman \(1993\)](#) who find no difference between the genders. We find no statistically precise relation between being single, having children, being full-time employed, living in a rural area, or age. Consumers with higher education are less knowledgeable about grocery prices, but even though we might expect higher income to reduce price consciousness ([Estelami et al. \(2001\)](#); [Estelami and Lehmann \(2001\)](#); [Wakefield and Inman \(1993\)](#); [Loy et al. \(2020\)](#)), high income is in our sample associated with slightly more price knowledge, which is also reflected in the raw correlation. Small and sometimes counterintuitive effects of age, income and employment on price knowledge are not new in the literature. [Gaston-Breton and Raghubir \(2012\)](#) points out that these sociodemographic variables are all correlated with both the ability and motivation to memorize prices in ways that can cancel out, leaving their relation to price knowledge theoretically ambiguous. For example, older consumers might have more time and incentives to think about grocery prices but also poorer memory. Similarly, high income or education can be correlated with both good memory and lower motivation to focus on grocery prices.

Consumer statements about shopping habits Price knowledge does not appear to be related to, all else being equal, the number of different grocery chains respondents regularly visit or whether they primarily shop at discount stores. However, direct and indirect markers of price consciousness do independently predict price knowledge; namely being the sole and primary grocery shopper of the household (as opposed to sharing or not having the main responsibility), being a member of all three main customer loyalty

¹⁹We also test how two-way clustering on municipality and product $\times$ chain combinations changes inference (see Table B.1 in Appendix B). This eliminates the significance, at the 5% level, of the indicators for supermarket and private label, the price coefficient of variation, and the time since last purchase. The latter remains significant in the fixed effects specification. However, there are only 48 clusters in the second dimension (and now 152 in the first), and persistency across recalled prices is likely stronger within individuals than within product $\times$ chain combinations. Since clustering on product $\times$ chain allows for arbitrary correlation across recalled prices on the same product-chain combination, our assessment is that doing so might be too conservative. Thus, we stick to individual-level clustering for our main specification, but remain careful in over-interpreting the coefficients that are affected.

programs (as opposed to only one or two).²⁰

Gathering information about grocery prices is strongly related to price knowledge: A one-point increase in the price learning effort index (see questions 4-7 in Table C.1 in Appendix C), which is less than the inter-quartile range, has a stronger association with price knowledge than either primary shopping responsibility or loyalty program membership. Price sensitivity, represented by questions 1-3 in Table C.1 in Appendix C, is almost completely unrelated to price knowledge, conditional on price learning effort.

The finding that price learning effort has an independent effect on price knowledge suggests that improving access to price information could be an effective way to raise consumer awareness. If knowledge were primarily driven by price sensitivity, shopping patterns, or demographics, such interventions would likely have limited effect. In the Norwegian grocery market, where there currently is no formal price transparency regulation or comprehensive comparison tools, collecting price information remains relatively costly for consumers. Policies that reduce these costs could both encourage more consumers to seek out prices and increase the overall level of price knowledge in the market.

Transaction data Turning now to the variables derived from the transaction data, we find that more frequent visits to the stores of the relevant chain is related to better price recall. This is consistent with what [Loy et al. \(2020\)](#) find using self-reported shopping frequencies. The weekly average number of unique products purchased at each chain, a variable that we are not aware of being used in the literature before, is negatively associated with price knowledge. One explanation for the negative association between basket variety and price knowledge could be that it is difficult to keep track of the prices of any given product if you tend to buy a large variety of products.

Price knowledge is decreasing in category size, defined by the number of close substitutes according to the grocery chain: Increasing category size by 10 units is associated with 3% less price recall accuracy. The findings are consistent with [Vanhuele and Dr  ze \(2002\)](#), who refer to the number of SKUs in a category as "clutter" and argue that increased category complexity increases the risk of confusion and poor recall.

Similarly, more price variation over time makes it harder to stay informed and predicts poorer price recall. An inter-quartile range increase in the price coefficient of variation (0.09) is associated with 1.1% less price recall accuracy. [Loy et al. \(2020\)](#) and [Vanhuele and Dr  ze \(2002\)](#) find a related result that more price variation within the product's category is associated with poorer recall. However, our long purchasing histories allow us to see how variation over time in the price of the product itself affects learning and recall, as opposed to the cross-sectional price variation within the product's category.

²⁰Note that these three programs cover more than 95% of the Norwegian grocery market. The primary tangible benefit of being a member of all the three main customer loyalty programs is access to personal price discounts, signaling price consciousness, and hence the estimated positive effect.

Having spent 1000 NOK more on the product, over the last 18 months, is associated with 4.3% better price recall. This is conditional on ever having bought the product and how long ago that was, so the effect of spending on price recall accuracy is not driven by familiarity with the product, but rather how important it is for the consumer to keep track of its price. In other words, consumers make sure to stay more informed about products for which the potential savings are the greatest. This is line with the predictions of [Stigler \(1961\)](#) concerning the incentives for search, i.e. price knowledge acquisition.

The respondents are much more informed about products that they have experience with buying: Recall accuracy is 7% higher on average for products that the consumer has purchased before. What is somewhat surprising, is that this effect seems to be very persistent. Even after 12 months, the effect associated with buying the product has only diminished with approximately $12 \times -0.001 = -1.2\%$. It seems that after a consumer buys a product the consumer continues to stay informed about the price of that product even if he or she buys it infrequently.

The marketing literature has long emphasized widespread consumer inattention and limited price knowledge but has provided less evidence on which consumers are more or less informed. By combining survey and transaction data, we show that price knowledge increases with the frequency of store visits, recent purchase experience, and spending on the product in question—suggesting that consumers acquire price knowledge through repeated exposure and focused attention. In contrast, broad or infrequent shopping reduces awareness. Although one strategy is to target promotions at already attentive shoppers, our findings also point to potential gains from directing pricing and communication efforts toward less frequent shoppers, who are otherwise unlikely to notice temporary price reductions but may be more responsive if made aware.

5 Concluding Comments

This paper provides the first direct evidence that consumers with better knowledge of grocery prices systematically pay lower prices. Using a unique dataset that links survey-based price recall measures to detailed, individual-level grocery purchase histories, we quantify the relationship between price knowledge and discount utilization. We show that consumers with more accurate knowledge of prices systematically pay lower prices, not by purchasing inherently cheaper products or frequenting cheaper stores, but by timing their purchases to take advantage of price variation over time. Our results also show that the benefits of price knowledge are economically meaningful: a 30-percentage-point improvement in price recall is associated with an average price reduction of about 4 percent.

Our findings provide direct empirical support for a central mechanism in search-theoretic models of sales—such as [Varian \(1980\)](#)—in which informed consumers impose

a competitive constraint on firms by responding strategically to price variation. By demonstrating that price knowledge is associated with paying lower prices—also after controlling for demographics and or self-reported price consciousness—we provide micro-level support for a key theoretical assumption that underlies a broad class of models in industrial organization and marketing.

Our finding on the price learning effort suggests that policies that are effective in increasing consumer price knowledge can help intensify competition and improve market efficiency. Examining the drivers of price knowledge, we find that consumers who are most active in gathering information about prices have higher price knowledge, and that knowledge seems to increase with price consciousness. Taken together our results support the idea that price disclosure regulations and other policies that lower search costs (by making it easier for consumers to learn prices) may raise price knowledge and, in turn, intensify competition. Since we are only considering consumer responses, one obvious caveat for such a conclusion is that we neglect potential supply responses from the firms, who will also learn more about competitor prices.

By linking survey-based recall data with transaction records, we provide a more detailed analysis of the determinants of price knowledge than survey data alone can offer. We find substantial heterogeneity between consumers, with greater price knowledge among those who visit stores frequently, have recently purchased the product, and spend more on it. In contrast, broad or infrequent shopping, larger product categories, and greater price variation are associated with lower knowledge. While attentive shoppers are more likely to benefit from temporary discounts, our findings suggest that targeted pricing and communication strategies aimed at less engaged consumers may offer untapped potential.

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Appendix

A Data Cleaning, Descriptive Statistics and Correlations

A.1 Data Cleaning

A few price recalls are so extreme that they are likely erroneous entries or at least not based on a serious attempt at recalling the right price. We filter out all individuals that make a price recall that is not within 10 and 300 percent of the true price. Furthermore, we exclude respondents that do not make at least 18 out of the 36 possible product recalls on the account of likely lack of focus and effort. This leaves us with 2028 respondents, 2 chains, and 24 products, and 72 392 observations, a loss of 11 % in terms of respondents and 7 % in terms of recalled prices.²¹

A.2 Descriptive Statistics

The primary outcome and explanatory variables that we use in the individual-level analysis are summarized in Table A.1, and the individual $\times$ chain $\times$ product-level, or recall-level, variables are summarized in Table A.2. We include the mean, standard deviation, minimum, maximum, and quartile values. It turns out that the variables that overlap between the tables have identical means, mainly because most respondents answer all 18 price questions, but the variables derived from the price recalls have different quantiles and lower standard deviations.

²¹Fewer observations end up being used in the regression analysis due to missing values for some of the survey variables. Finally, we also checked for respondents with very few transactions in their purchasing histories, but removing these respondents does not meaningfully alter the results.

Table A.1: Summary statistics: individual-level variables

	Mean	SD	Min	P25	Median	P75	Max
<i>Panel A: Discount Utilization and Purchase-Based Outcomes</i>							
Discount Utilization (Simple)	0.040	0.037	−0.044	0.013	0.033	0.059	0.272
Discount Utilization (Sales-Weighted)	0.021	0.027	−0.073	0.004	0.014	0.031	0.219
<i>By reference-price window (simple):</i>							
Daily	0.002	0.015	−0.136	−0.002	0.000	0.006	0.221
1 Week	0.007	0.017	−0.054	−0.001	0.002	0.011	0.229
2 Weeks	0.015	0.022	−0.050	0.001	0.010	0.023	0.232
Monthly	0.027	0.031	−0.059	0.006	0.020	0.041	0.262
6 Months	0.043	0.040	−0.048	0.015	0.036	0.064	0.265
Full Period	0.044	0.043	−0.115	0.014	0.037	0.067	0.262
Discount Utilization (Quarterly, Timing Only)	0.037	0.031	−0.037	0.014	0.032	0.055	0.186
<i>Share of purchases on sale:</i>							
≥3%	0.158	0.079	0	0.104	0.142	0.191	0.755
≥5%	0.144	0.077	0	0.094	0.128	0.174	0.735
≥15%	0.110	0.071	0	0.063	0.092	0.139	0.714
≥25%	0.082	0.058	0	0.042	0.067	0.106	0.551
Basket Coefficient of Variation (Simple)	0.11	0.01	0.07	0.10	0.11	0.12	0.17
Basket Coefficient of Variation (Sales-Weighted)	0.10	0.02	0.04	0.08	0.09	0.11	0.16
<i>Panel B: Price Knowledge</i>							
Price Knowledge (PK)	−0.27	0.08	−0.70	−0.32	−0.27	−0.22	−0.04
Price Bias (PD)	0.02	0.13	−0.44	−0.06	0.01	0.09	0.60
Share Over-estimated	0.46	0.15	0	0.36	0.44	0.56	0.92
Share Under-estimated	0.53	0.15	0.06	0.44	0.53	0.64	1
Share Correct	0.01	0.04	0	0	0	0	0.53
Share Within 1%	0.06	0.07	0	0.03	0.06	0.08	0.56
Share Within 5%	0.17	0.11	0	0.08	0.14	0.22	0.78
Share Within 10%	0.29	0.13	0	0.19	0.28	0.36	0.89
Share Within 20%	0.50	0.15	0.06	0.40	0.50	0.58	1
<i>Panel C: Demographics</i>							
Male	0.38	0.49	0	0	0	1	1
Single Person Household	0.12	0.33	0	0	0	0	1
Has Children	0.42	0.49	0	0	0	1	1
Full-Time Employment	0.61	0.49	0	0	1	1	1
Higher Education	0.71	0.45	0	0	1	1	1
Rural	0.30	0.46	0	0	0	1	1
Household Income > 950' NOK	0.54	0.50	0	0	1	1	1
Age: Below 40 Years	0.23	0.42	0	0	0	0	1
Age: 40-59 Years	0.47	0.50	0	0	0	1	1
Age: Above 60 Years	0.30	0.46	0	0	0	1	1
<i>Panel D: Shopping Behavior and Attitudes</i>							
# Frequented Grocery Chains	2.98	1.39	1	2	3	4	9
Mainly uses discount stores	0.86	0.35	0	1	1	1	1
Average Weekly Store Visits	2.90	1.80	0.09	1.69	2.58	3.65	16.53
Main Grocery Shopper	0.74	0.44	0	0	1	1	1
Average Weekly Variety	27.98	13.23	1.39	18.80	26.15	35.09	102.72
Three Customer Membership Programs	0.48	0.50	0	0	0	1	1
Price Consciousness	3.81	0.75	1	3.36	3.91	4.38	5
Price Sensitivity	4.12	0.79	1	3.74	4.26	4.74	5

Table A.2: Summary statistics: individual $\times$ chain $\times$ product-level variables

	Mean	SD	Min	P25	Median	P75	Max
<i>Panel A: Price Knowledge</i>							
Price Knowledge (PK)	-0.27	0.26	-1.99	-0.38	-0.20	-0.08	0
Price Bias (PD)	0.02	0.37	-0.90	-0.22	-0.03	0.17	1.99
Share Over-estimated	0.46	0.50	0	0	0	1	1
Share Under-estimated	0.53	0.50	0	0	1	1	1
Share Correct	0.01	0.11	0	0	0	0	1
Share Within 1%	0.06	0.24	0	0	0	0	1
Share Within 5%	0.17	0.37	0	0	0	0	1
Share Within 10%	0.29	0.45	0	0	0	1	1
Share Within 20%	0.50	0.50	0	0	1	1	1
<i>Panel B: Demographics</i>							
Male	0.38	0.49	0	0	0	1	1
Single Person Household	0.12	0.33	0	0	0	0	1
Has Children	0.42	0.49	0	0	0	1	1
Full-Time Employment	0.61	0.49	0	0	1	1	1
Higher Education	0.71	0.45	0	0	1	1	1
Rural	0.30	0.46	0	0	0	1	1
Household Income > 950' NOK	0.54	0.50	0	0	1	1	1
Age: Below 40 Years	0.23	0.42	0	0	0	0	1
Age: 40-59 Years	0.47	0.50	0	0	0	1	1
Age: Above 60 Years	0.30	0.46	0	0	0	1	1
<i>Panel C: Shopping Behavior and Attitudes</i>							
# Frequented Grocery Chains	2.98	1.39	1	2	3	4	9
Mainly uses discount stores	0.86	0.35	0	1	1	1	1
Main Grocery Shopper	0.74	0.44	0	0	1	1	1
Three Customer Membership Programs	0.48	0.50	0	0	0	1	1
Price Consciousness	3.81	0.75	1	3.36	3.91	4.38	5
Price Sensitivity	4.12	0.79	1	3.74	4.26	4.74	5
Price Learning Effort	3.52	0.89	1	2.95	3.62	4.22	5
<i>Panel D: Chain, Product, and Purchase Characteristics</i>							
Supermarket Chain	0.50	0.50	0	0	0.50	1	1
Private Label	0.08	0.28	0	0	0	0	1
Average Weekly Chain Visits	1.45	1.56	0	0.23	0.99	2.21	15.56
Average Weekly Chain Variety	13.99	14.37	0	1.42	9.67	23.01	89.13
Category Size	19.34	19.11	2	7	12.92	25	128
Price Coefficient of Variation	0.11	0.06	0.01	0.06	0.11	0.15	0.33
Sales	92.33	363.26	0	0	0	31.56	18492.96
Has Purchased Product at Chain	0.31	0.46	0	0	0	1	1
Months Since Last Purchase	4.22	4.22	0	0.73	0.73	5.56	15.56

Table A.2 shows that the mean PK of the recalled prices is -0.27, meaning that the average price recall either undershoots or overshoots the true price by 27%. In addition to the PK defined in Equation 2.1 above, we include the share of recalled prices within different error bounds: $e = \{0, 0.01, 0.05, 0.1, 0.2\}$ defined in Equation ???. Only 1% of the product recalls are exactly correct at the two-decimal level, i.e. approximately 724 out of the 72 392 recalled prices.

Loy et al. (2020), which surveyed grocery shoppers in two stores in Northern Germany, found that the average PK in their sample is 35% and that around 14% of recalled prices lay within 5% of the true price. This is a slightly poorer performance than our sample with an average PK of 27% and a share of recalled prices within 5% of the true price of 17%. The magnitudes are fairly similar, but of course, the results are sensitive to the products asked about (see Figure C.2a), the population surveyed, and the setting in which the respondents were surveyed (see Evanschitzky et al. (2004) for a discussion).

The price recalls overestimate the correct prices by 2% on average. Consistent with the findings of ?, which measured price knowledge among Finnish grocery shoppers, the bias of the respondents as a group is low, despite substantial errors on any given recall. However, this breaks down somewhat at the chain and product level (see Figures C.1b and C.2b in the Appendix). Although respondents only slightly underestimate the prices of the discount chain on average ($PD = -1.5\%$), they overestimate the prices of the supermarket chain by 5.2%. This likely reflects the considerable efforts that discount chains in Norway, who serve over 60% of the market, put into building and maintaining a favorable price image. Supermarkets, on the other hand, focus more on service, assortment size, and quality. Furthermore, the price bias varies substantially across products. For example, the respondents are far off target, on average, for strawberry jam (60%), crisp bread (49%), potato chips (33%), and frozen cod (-40%), whereas the bias is quite low for eggs (-2%), smoothies (-2%), and 10 other products with less than 10% of price bias.

The share of overestimated recalls is 46%, and the share of underestimated recalls is 53%, despite the average price bias being positive. This stems from the overestimated price recalls being larger in magnitude (see Figure C.2b in Appendix C), which is also reflected in the median price bias (-3%) being lower than the average (2%). The shares of overestimation and underestimation differ substantially from the findings of Loy et al. (2020), who find that 56% of price recalls overestimate the true price, and only 36% underestimate true prices. However, they find that 8% of recalls are exact, which we suspect can be explained by differences in the number of decimal places accounted for. We do not know if some of the 8% would be counted as over or under-estimations in our study. However, if we allow for slight errors within 1%, we get a success rate of 6%, which is much closer to the results of Loy et al. (2020).

The explanatory variables that we use in the main regression analyses are also included in Tables A.1 and A.2. To simplify the analysis, we mostly use binary indicators derived

from the detailed questions included in the survey. Both tables show that the sample is quite representative in terms of demographics, although compared to the general population, more respondents have higher education (71%) and are older than 40 years old (77%), and there are more female than male respondents (38%). From the averages of price consciousness (3.81), sensitivity (4.12), and learning effort (3.52), we understand that most respondents agree or strongly agree on statements related to price sensitivity, whereas respondents are more mixed on activities that generate price knowledge (see Figure C.5).

The variables that are unique to Table A.1 are those related to discount utilization and shopping basket price variation, which are calculated from the full purchasing histories of the respondents, not just the products included in the survey. We see that most consumers have positive discount utilization, meaning that they do better than shopping at random. Furthermore, simple discount utilization is higher across the quartiles than sales-weighted discount utilization, which suggests that the products that consumers spend the most on are not the products they buy the most at discounted prices, despite being incentivized to look for sales on products that they can make the most savings on. One explanation is that the products people spend the most on have fewer and smaller discounts for consumers to act on. This is supported by the lower average price coefficient of variation for the products that the respondents buy, at 0.11, compared to the price variation of their average shopping baskets, at 0.1, which tells us that the products that the respondents spend more on vary less in price over the sample period.

Table A.2 includes variables that are unique to the recall-level analysis. The share of price recalls for the supermarket and the discount chain is even, and 8% of recalls were on private label products (2 out of the 24 included products). The average weekly number of visits per chain is 1.45 and the average weekly number of unique products purchased at a given chain is 14. The average recalled price is related to a product that has 19 close substitutes, defined by the category it belongs to, and that has an average price coefficient of variation, now measured for a specific product $\times$ chain-combination included in the survey, of 0.11, which happen to be identical to the simple average across all products that the respondents purchase. This indicates that the list of products that we include in the survey are representative of all products that consumers buy in terms of price variation.

Out of all the recalled prices that are included in the analysis, 31% were given by respondents that at some point purchased the product at the relevant chain during the sample period. Among those, the average time since purchase is 4.7 months whereas the median is 2.7 months. This might suggest that most respondents buy the same products more frequently, but that a few respondents go much longer between purchases or are even one-time buyers, pulling up the average. The average sales spent on the product being recalled during the sample period of 18 months is 92 NOK, but this includes the 69% of recalls for products on which there is zero spending, which is reflected by the

median spending of 0 NOK. The total spending distribution among those that bought the product is also skewed, with the average and 3rd quartile spending both being 300 NOK, the 95th-percentile being 1200 NOK, and the observed maximum being 18500 NOK.

A.2.1 Correlations

We implement several markers of price consciousness in the analysis, the correlations of which are found in Figure A.1. If they are all to reliably capture some aspect of price consciousness, we should expect them to be correlated. In particular, it is interesting to see if the price consciousness index based on subjective assessments coincide with markers of price consciousness that are revealed through behavior. Figure A.1 shows that the price consciousness index is moderately correlated with the other measures in the ways we would expect, although the relation between the index and household shopping responsibility is weaker ($\rho_s = 0.05$). It is reassuring that discount utilization and the share of spending on low-cost private labels, both revealed from the transaction data, are the most correlated with the price consciousness index with coefficients of $\rho_s = 0.32$ and $\rho_s = 0.33$, respectively.

The patterns are similar when we look at price sensitivity and learning effort separately, except that mainly using discount stores is more related to price sensitivity and discount utilization is more related to price learning effort. Taken together, these findings indicate that stated and revealed markers of price consciousness are significantly related, but also that one cannot perfectly predict the other. We will therefore rely on both in our analysis.

Figure A.2 contains the correlations between all the variables used in the individual-level analysis. Many variables are not very correlated, but things like having children, being full-time employed, income, and age are strongly correlated in ways one would expect. Mainly using discount stores is correlated with less discount utilization and less price variation in the products you buy (consistent with less sales activity), and younger age. Men are less likely to be the main grocery shopper of the household, and buying many different products per week is strongly related to frequent store visits. Finally, frequent store visits is related to more discount utilization, consistent with more opportunities to act on sales.

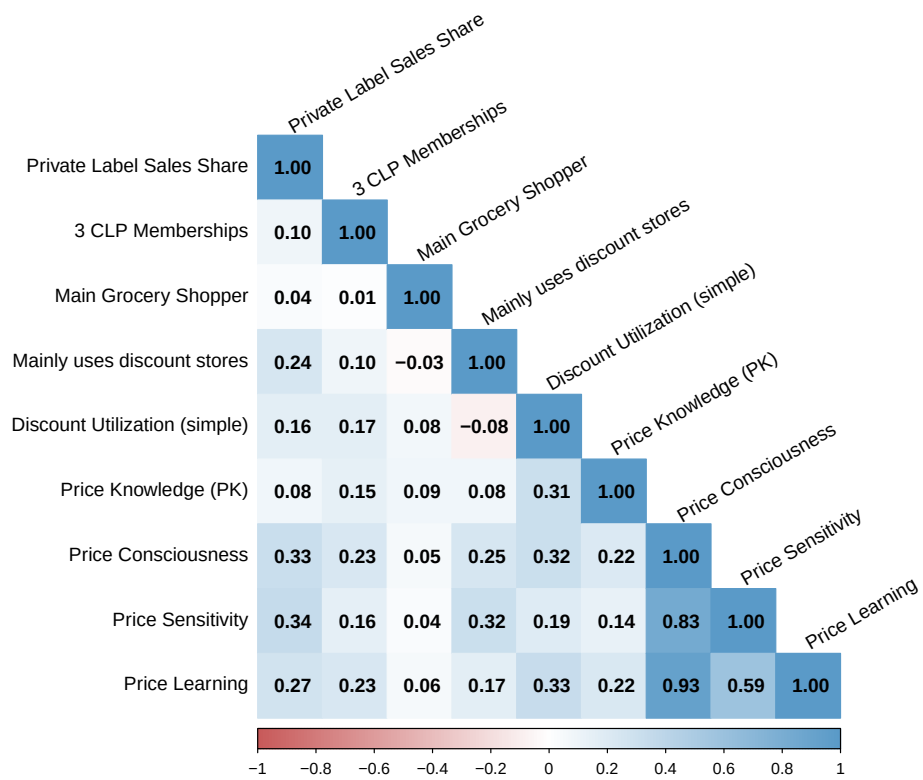


Figure A.1: Each cell contains the *Spearman's rank correlation coefficient* for markers of price consciousness and price knowledge.

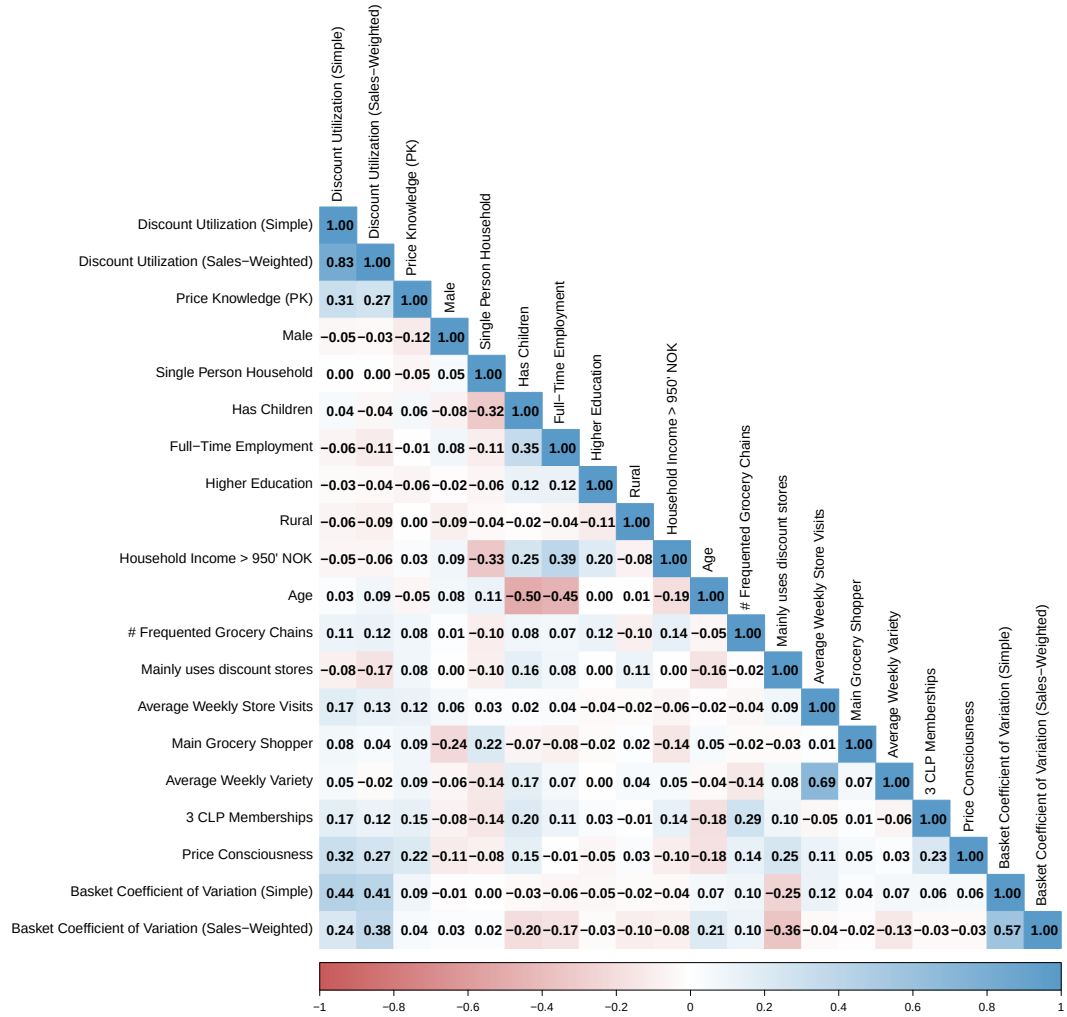


Figure A.2: Each cell contains the *Spearman's rank correlation coefficient* for individual-level outcome and explanatory variables based on survey responses and the respondents' purchasing history within the categories covered by the survey.

Figure A.3 focuses on the variables that are unique to the recall-level analysis. It is clear that the respondents in the sample visit and buy more products from the discount chain compared to the supermarket chain, and that store visits and the number of products are strongly correlated. Products have more close substitutes at the supermarket chain, the private label products have fewer close substitutes overall, and the products that do have many close substitutes tend to vary a lot in price as well. Naturally, sales and the indicator for ever purchasing the product is strongly related, especially based on monotonic correlation. Similarly, higher spending is associated with more recent purchases, and store visiting frequency goes hand in hand with more sales and purchases, which explains lower sales and purchases being associated with the supermarket chain.

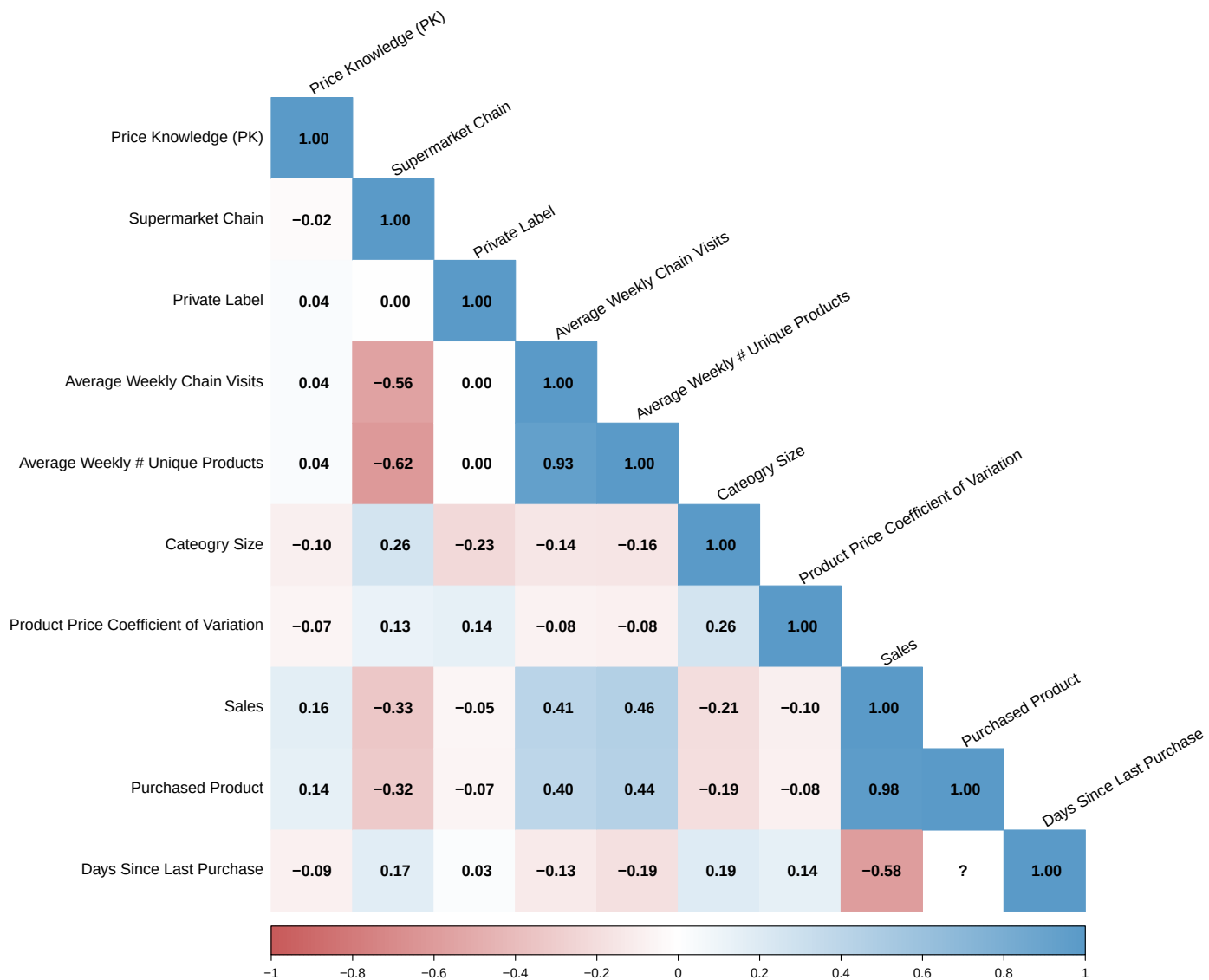


Figure A.3: Each cell contains the *Spearman's rank correlation coefficient* for respondent x chain x product-level outcome and explanatory variables based on survey responses and the respondents' purchasing history within the categories covered by the survey.

B Robustness Checks and Extensions

B.1 Two-Way Clustering

Table B.1 replicates the results in Table 4.1 using standard errors clustered on home municipality and product $\times$ chain-combinations. This allows for correlation between prices recalled by consumers living in the same municipality, including their own recalled prices, and persistence in the way different consumers recall the prices of specific product $\times$ chain-combinations. The most important correlation likely stems from a consumer that is well informed about one product being more likely to be informed about prices in general, but it could also be the case that people living in the same municipality face similar price cues. Similarly, certain products might be promoted in such a way that consumers share the same biases.

The statistical significance of most coefficients are practically unchanged, but the coefficients on supermarket chain, private label, price coefficient of variation, and months since last purchase in Column (3) are no longer statistically significant. Except for the latter, which is still significant in the fixed-effects specification, these variables are features of the product $\times$ chain-clusters which now are allowed to correlate in arbitrary ways across price recalls on a given product $\times$ chain-combination. In other words, the respondents' ability to recall the prices of the same chain-specific product can be anything from perfectly correlated to i.i.d. This constrains our ability to learn much about these variables, which is reflected in the large standard errors of their coefficients. It is unlikely that respondents recall prices almost equally well for a given product $\times$ chain-combination, regardless of promotions or sales, so these standard errors are likely conservative.

Table B.1: Price knowledge regressions: two-way clustering

	Price Knowledge (PK)		
	(1)	(2)	(3)
Male	-0.015** (0.005)	-0.015** (0.005)	-0.015** (0.005)
Single Person Household	-0.002 (0.008)	-0.002 (0.008)	-0.001 (0.008)
Has Children	0.002 (0.005)	0.003 (0.005)	-0.001 (0.005)
Full-Time Employment	-0.005 (0.004)	-0.005 (0.004)	-0.004 (0.004)
Higher Education	-0.013** (0.004)	-0.013** (0.004)	-0.012* (0.004)
Rural	0.000 (0.004)	0.000 (0.004)	-0.002 (0.005)
Household Income > 950' NOK	0.010* (0.004)	0.010* (0.004)	0.008* (0.004)
Age: 40-59 Years	0.008 (0.006)	0.008 (0.006)	0.006 (0.005)
Age: Above 60 Years	0.003 (0.010)	0.004 (0.010)	0.002 (0.009)
# Frequented Grocery Chains	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)
Mainly Uses Discount Stores	0.004 (0.006)	0.006 (0.006)	0.004 (0.005)
Main Grocery Shopper	0.014** (0.005)	0.013** (0.005)	0.014** (0.005)
Supermarket Chain	-0.017 (0.011)	-0.017 (0.011)	0.033 (0.020)
Private Label	0.044 (0.047)	0.044 (0.047)	0.018 (0.038)
3 Customer Loyalty Program Memberships	0.011** (0.004)	0.011* (0.004)	0.010* (0.004)
Price Consciousness	0.019*** (0.003)		
Price Sensitivity		0.001 (0.003)	0.000 (0.003)
Price Learning Effort		0.016*** (0.002)	0.015*** (0.002)
Average Weekly Chain Visits			0.006** (0.002)
Average Weekly Chain Variety (per 10 products)			-0.010*** (0.002)
Category Size (per 10 products)			-0.029 (0.014)
Price Coefficient of Variation			-0.120 (0.277)
Purchased			0.068** (0.021)
Purchased $\times$ <i>Sales</i> in 1000 NOK			0.043*** (0.008)
Purchased $\times$ <i>Months Since Last Purchase</i>			-0.001 (0.001)
Constant	-0.354*** (0.030)	-0.344*** (0.029)	-0.308*** (0.034)
Observations	69,408	69,408	69,408
Dependent variable mean	-0.27	-0.27	-0.27
R ²	0.01	0.01	0.08

Notes: This table reports results from estimation of Equation ???. Price knowledge (PK) is calculated as the negative percentage absolute deviation between the respondents' recalled price and the true shelf price. Price consciousness is calculated as the factor-weighted average of seven Likert-scale survey items regarding price sensitivity and learning effort, while price sensitivity and learning effort are calculated similarly using Likert items 1–3 and 4–7, respectively. The average weekly chain variety is calculated as the average number of unique products purchased at each chain per week over the sample period. Category size is the chain-specific number of close substitutes for the product as defined by the grocery chain. The price coefficient of variation is calculated for each product $\times$ chain combination over the sample period. Purchased is an indicator of whether the respondent bought the product at a given chain during the sample period, and sales is the consumer's total product-chain spending. Column 1 includes individual characteristics and indicators for private labels and supermarket chain, Column 2 splits price consciousness into sensitivity and learning effort, and Column 3 adds variables derived from transaction data. The standard errors shown in parentheses are clustered by individuals and products. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

B.2 Discount Utilization and The Share of Correct Price Recalls

Table B.2 shows the OLS results of a simple regression of discount utilization on the share of recalled prices within some error bound, using the sample including outliers in PK, for comparison with the results in Table 3.1:

$$\text{DU}(Q)_i^{\text{Simple}} = \alpha + \delta \cdot W_i^e + \epsilon_i \quad (\text{B.1})$$

where W_i^e is the share of correctly recalled prices within some error bound e . Like a high PK, a high share of correctly recalled prices is predictive of more discount utilization. This is reassuring, since the share of correct recalls is robust to extreme price recalls, for example from erroneous entry or inattention, that could significantly affect the PK. However, the explained variance is larger for the share of roughly correct recalls, i.e. within 20%. Seemingly, precise knowledge of prices is not needed in order to act on sales — a rough idea is enough. Still, the PK performs slightly better in explaining discount utilization with an R^2 of 7% compared to the R^2 of 5% obtained by using the share of roughly correct recalls.²² Finally, note that the larger coefficients on precise compared to rough recalls does not signify that precise recalls are more important, since a given increase in the share of precise recalls is much harder to obtain than an equal increase in the share of roughly correct recalls.

²²This also holds when comparing like-for-like on the sample without outliers.

Table B.2: Discount utilization regressions: within percentage error bounds

	Discount Utilization			
	(1)	(2)	(3)	(4)
Share Within 1%	0.064*** (0.012)			
Share Within 5%		0.060*** (0.007)		
Share Within 10%			0.051*** (0.006)	
Share Within 20%				0.051*** (0.005)
Constant	0.035*** (0.001)	0.030*** (0.001)	0.025*** (0.002)	0.014*** (0.002)
Observations	2,285	2,285	2,285	2,285
Dependent variable mean	0.04	0.04	0.04	0.04
R ²	0.02	0.03	0.04	0.05

Notes: This table reports results from estimation of discount utilization regressions using the share of correct price recalls within different error bounds and the sample including outliers in price knowledge. Discount utilization is calculated as the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product $\times$ chain combinations using simple weights. Columns (1-4) uses error bounds of 1, 5, 10, and 20 percent as explanatory variables, respectively. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

B.3 Explaining The Likelihood of Correct Price Recalls

Table B.3 contains maximum likelihood results from logistic regressions of indicators for correct price recalls, within different error bounds, on the same explanatory variables as in Column 3 in Table 4.1, based on price recalls including those that are extreme outliers:

$$Pr(W_{ijc}^e = 1 | \{\theta, \mathbf{X}_{ijc}\}) = \Lambda(\theta + \gamma' \mathbf{X}_{ijc}) = \frac{1}{1 + e^{-(\theta + \gamma' \mathbf{X}_{ijc})}} \quad (\text{B.2})$$

By and large, the results mirror the results in column 3 in Table 4.1 in terms of signs and statistical significance, especially for the more lenient error bounds. This is reassuring since the indicators for correct recalls are immune to outliers, and suggests that our main results are not dependent on our criteria for removing outliers.

The results in Table B.3 also allow us to see how the explanatory variables differ in explaining precisely correct vs. roughly correct price recalls. In most cases, the average marginal effect on the probability of correct recall have the same sign across specifications, but is larger in magnitude for the regressions using the more lenient error bounds, reflecting greater difficulty in inducing precise recalls compared to roughly correct recalls.²³ However, while the ability to very precisely recall the correct price, i.e. within 1%, is diminished as the time since the last purchase increases, such an effect is not statistically detected for rough recalls, i.e. within 20%. Among those that have purchased the product, the probability of precise recall is reduced by 2.6% (from 9.1% to 6.5%) after 12 months, whereas the probability of roughly correct recall is only reduced by 1% (from 59% to 58%). This phenomenon might reflect exact prices being committed to short-term memory whereas only a rough notion of the price level remains in long-term memory, which is consistent with [Vanhuele and Dr ze \(2002\)](#) who show that consumers can be able to recognize deals even if they are not able to precisely recall the correct price.

²³This is also not surprising given the sigmoid shape of the logistic function and precise recalls being very rare compared to rough recalls.

Table B.3: Logistic price knowledge regressions: correct price recalls

	Pr(Within 1%) (1)	Pr(Within 5%) (2)	Pr(Within 10%) (3)	Pr(Within 20%) (4)
Male	-0.118* (0.055) [-0.7%]	-0.134*** (0.036) [-1.8%]	-0.131*** (0.030) [-2.6%]	-0.133*** (0.028) [-3.2%]
Single Person Household	0.094 (0.104) [0.5%]	-0.002 (0.064) [0.0%]	-0.032 (0.053) [-0.6%]	-0.037 (0.047) [-0.9%]
Has Children	0.000 (0.062) [0.0%]	0.033 (0.042) [0.4%]	0.017 (0.035) [0.3%]	0.005 (0.033) [0.1%]
Full-Time Employment	-0.040 (0.072) [-0.2%]	-0.007 (0.045) [-0.1%]	-0.039 (0.038) [-0.8%]	-0.028 (0.034) [-0.7%]
Higher Education	-0.183** (0.061) [-1.0%]	-0.172*** (0.039) [-2.4%]	-0.135*** (0.033) [-2.7%]	-0.092** (0.031) [-2.2%]
Rural	0.003 (0.059) [0.0%]	-0.025 (0.038) [-0.3%]	-0.013 (0.032) [-0.3%]	-0.010 (0.030) [-0.2%]
Household Income > 950' NOK	0.060 (0.064) [0.3%]	0.083* (0.041) [1.1%]	0.068* (0.034) [1.4%]	0.076* (0.031) [1.8%]
Age: 40-59 Years	0.061 (0.063) [0.3%]	0.030 (0.040) [0.4%]	-0.002 (0.034) [0.0%]	-0.023 (0.032) [-0.6%]
Age: Above 60 Years	-0.083 (0.100) [-0.5%]	-0.039 (0.062) [-0.5%]	-0.104* (0.052) [-2.0%]	-0.108* (0.047) [-2.6%]
# Frequented Grocery Chains	0.015 (0.019) [0.1%]	0.014 (0.012) [0.2%]	0.014 (0.010) [0.3%]	0.009 (0.010) [0.2%]
Mainly Uses Discount Stores	-0.102 (0.085) [-0.6%]	-0.005 (0.054) [-0.1%]	0.009 (0.046) [0.2%]	0.014 (0.042) [0.3%]
Main Grocery Shopper	0.067 (0.053) [0.4%]	0.069 (0.037) [0.9%]	0.074* (0.032) [1.5%]	0.098** (0.030) [2.4%]
Supermarket Chain	-0.132* (0.059) [-0.7%]	0.035 (0.028) [0.5%]	0.118*** (0.019) [2.3%]	0.200*** (0.016) [4.8%]
Private Label	-1.182*** (0.107) [-4.3%]	0.200*** (0.039) [2.8%]	0.128*** (0.033) [2.6%]	0.118*** (0.034) [2.8%]
Three Customer Program Memberships	0.110* (0.052) [0.6%]	0.055 (0.034) [0.7%]	0.084** (0.029) [1.7%]	0.084** (0.027) [2.0%]
Price Sensitivity	-0.061 (0.045) [-0.3%]	-0.024 (0.029) [-0.3%]	-0.019 (0.023) [-0.4%]	-0.013 (0.020) [-0.3%]
Price Learning Effort	0.126*** (0.037) [0.7%]	0.122*** (0.023) [1.6%]	0.130*** (0.020) [2.6%]	0.135*** (0.018) [3.2%]
Average Weekly Chain Visits	0.075* (0.031) [0.4%]	0.055** (0.017) [0.7%]	0.051*** (0.013) [1.0%]	0.054*** (0.012) [1.3%]
Average Weekly Chain Variety (per 10 products)	-0.075* (0.031) [-0.4%]	-0.069*** (0.018) [-0.9%]	-0.077*** (0.014) [-1.5%]	-0.091*** (0.012) [-2.2%]
Category Size	-0.062*** (0.011) [-0.3%]	-0.064*** (0.007) [-0.9%]	-0.049*** (0.005) [-1.0%]	-0.093*** (0.005) [-2.2%]
Price Coefficient of Variation	-1.381*** (0.305) [-7.7%]	-1.245*** (0.197) [-16.6%]	-0.766*** (0.172) [-15.3%]	-1.251*** (0.163) [-30.0%]
Purchased	0.417*** (0.049) [2.4%]	0.465*** (0.034) [6.6%]	0.418*** (0.030) [9.2%]	0.455*** (0.030) [12.2%]
Purchased $\times$ <i>Sales</i> in 1000 NOK	0.303*** (0.038) [2.2%]	0.278*** (0.034) [4.5%]	0.449*** (0.037) [10.0%]	0.588*** (0.049) [13.8%]
Purchased $\times$ <i>Months Since Last Purchase</i>	-0.030*** (0.006) [-0.2%]	-0.020*** (0.004) [-0.3%]	-0.009** (0.003) [-0.2%]	-0.004 (0.003) [-0.1%]
Constant	-2.630*** (0.198)	-1.888*** (0.130)	-1.304*** (0.112)	-0.377*** (0.100)
Observations	73,548	73,548	73,548	73,548
Baseline Probability	6%	16%	29%	49%
Pseudo R ²	0.03	0.02	0.02	0.03

Notes: This table reports results from estimation of Equation B.2 using maximum likelihood. Price knowledge (within) is an indicator for a recalled price being within a percentage error bound of the true shelf price. Price sensitivity and learning effort are calculated as the factor-weighted average of Likert items 1–3 and 4–7, respectively. The average weekly chain variety is calculated as the average number of unique products purchased at each chain per week over the sample period. Category size is the chain-specific number of close substitutes for the product as defined by the grocery chain. The price coefficient of variation is calculated for each product $\times$ chain combination over the sample period. Purchased is an indicator of whether the respondent bought the product at a given chain during the sample period, and sales is the consumer's total product-chain spending. Columns 1–4 parallel Column 3 in Table 4.1 using error bounds of 1, 5, 10, and 20 percent, respectively. The standard errors shown in parentheses are clustered by individual, and the average marginal effects are shown in square brackets. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

C Additional Tables & Figures

Table C.1: Factor loadings and index weights for price consciousness, price sensitivity, and price learning effort

Likert Scale Question	Price Consciousness		Price Sensitivity		Price Learning Effort	
	Loading	Weight	Loading	Weight	Loading	Weight
1. Price is crucial for my choice of products.	0.73	15.4%	0.883	39.7%	–	–
2. Price is crucial for my choice of store.	0.719	15.2%	0.771	34.7%	–	–
3. I buy on Sales and promotions as often as I can.	0.74	15.7%	0.571	25.7%	–	–
4. I usually compare prices between the grocery chains.	0.81	17.1%	–	–	0.86	32.9%
5. I usually check the grocery chains' flyers for prices.	0.717	15.2%	–	–	0.762	29.1%
6. I usually check the results of VGs grocery price comparison tests.	0.387	8.2%	–	–	0.411	15.7%
7. I usually check the shelf price of the product before I buy it.	0.624	13.2%	–	–	0.584	22.3%
Sum	4.727	100%	2.225	100%	2.617	100%

Notes: This table shows estimated factor loadings for each Likert scale item using diagonally weighted least squares, as implemented by the *lavaan* package in *R*. The weights put on each likert scale question in the indices for price consciousness, sensitivity, and learning effort are calculated by dividing the factor loadings by the sum of the factor loadings.

Comparison of Mean Price Error and Bias Between Grocery Chains

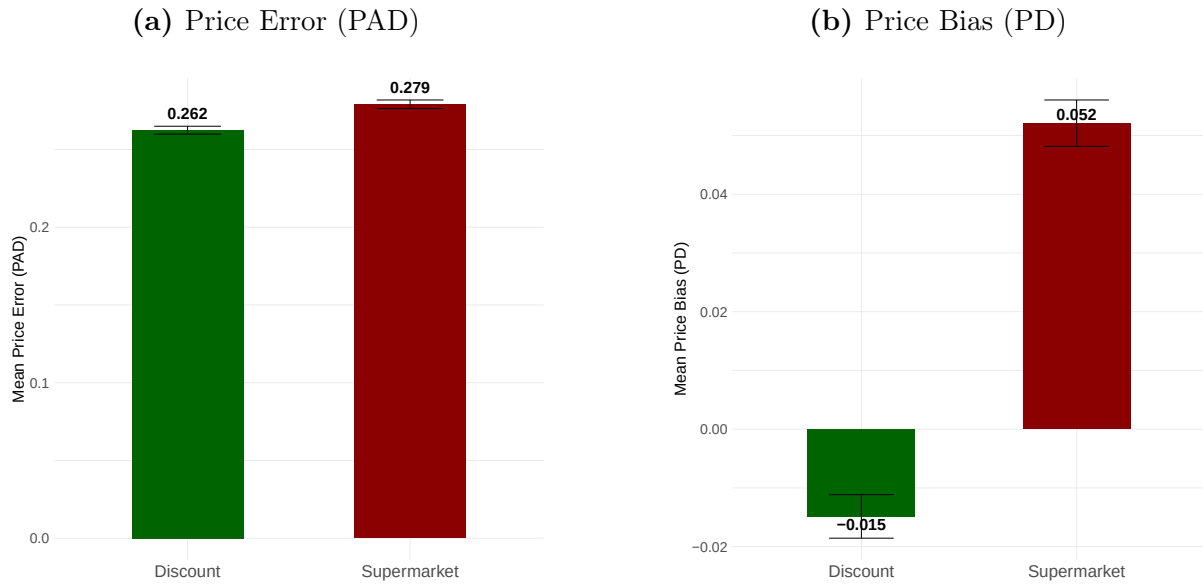
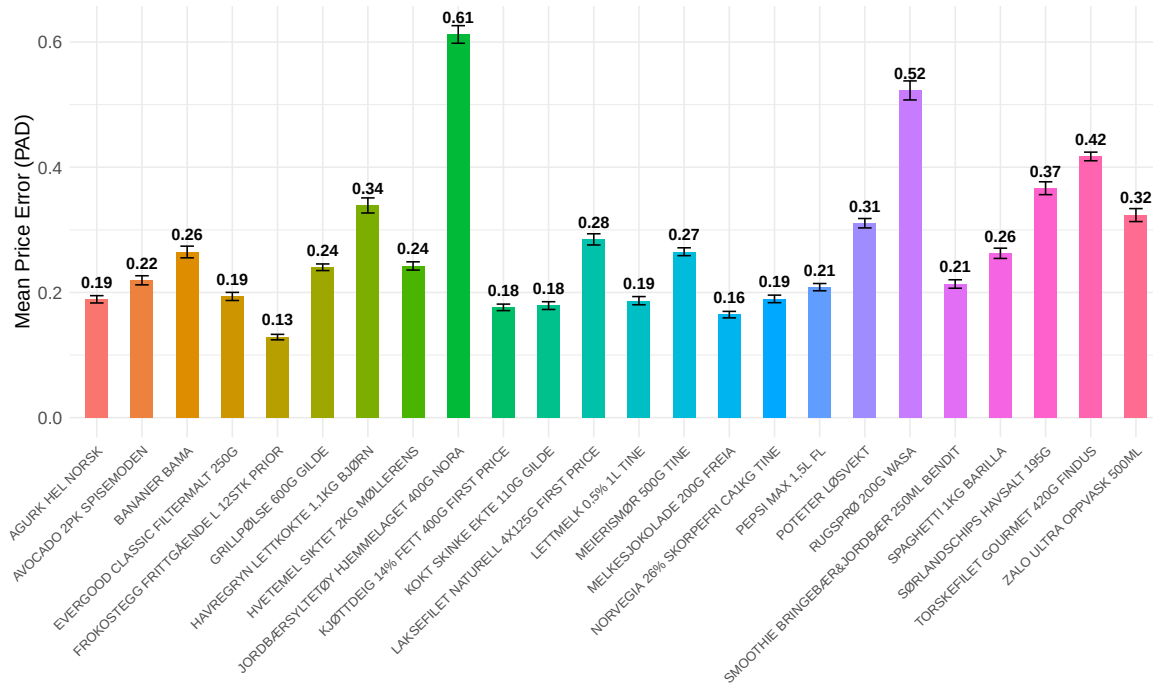


Figure C.1: Price error is measured as the percentage absolute deviation between the respondent's recalled price and the true price. Price bias is measured as the percentage deviation between the respondent's recalled price and the true price. Confidence intervals at 95 % are calculated using the Student's t-distribution.

Comparison of Mean Price Error and Bias Between Products

(a) Price Error (PAD)



(b) Price Bias (PD)

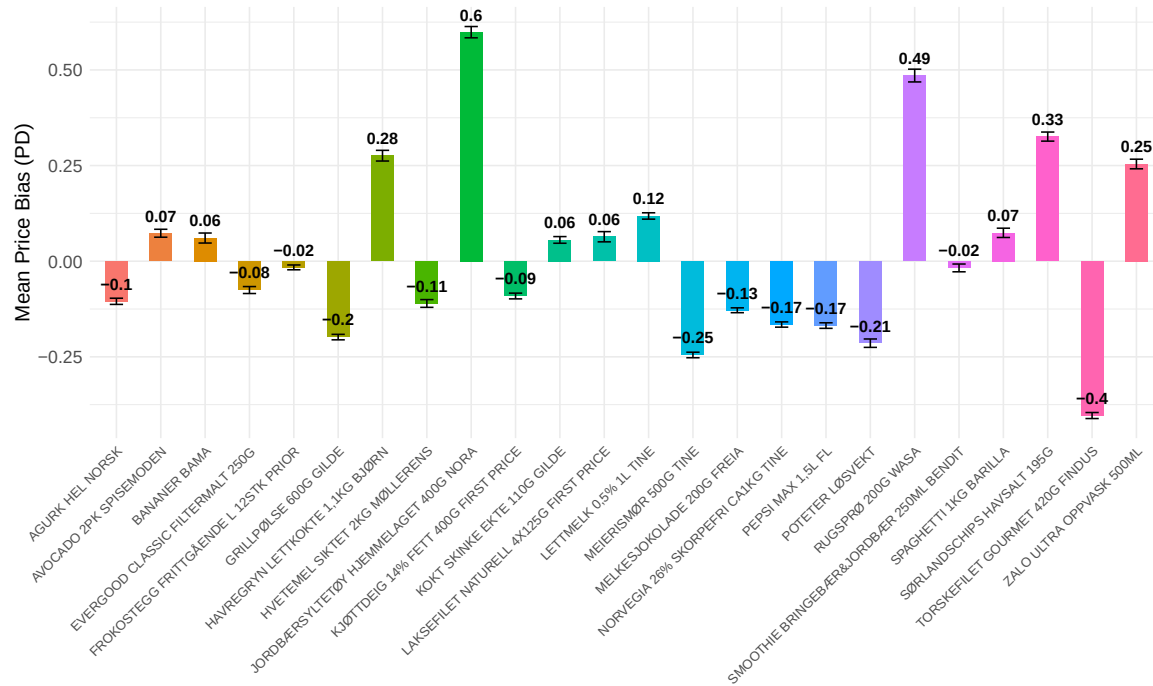


Figure C.2: Price error is measured as the percentage absolute deviation between the respondent's recalled price and the true price. Price bias is measured as the percentage deviation between the respondent's recalled price and the true price. Confidence intervals at 95 % are calculated using the Student's t-distribution.

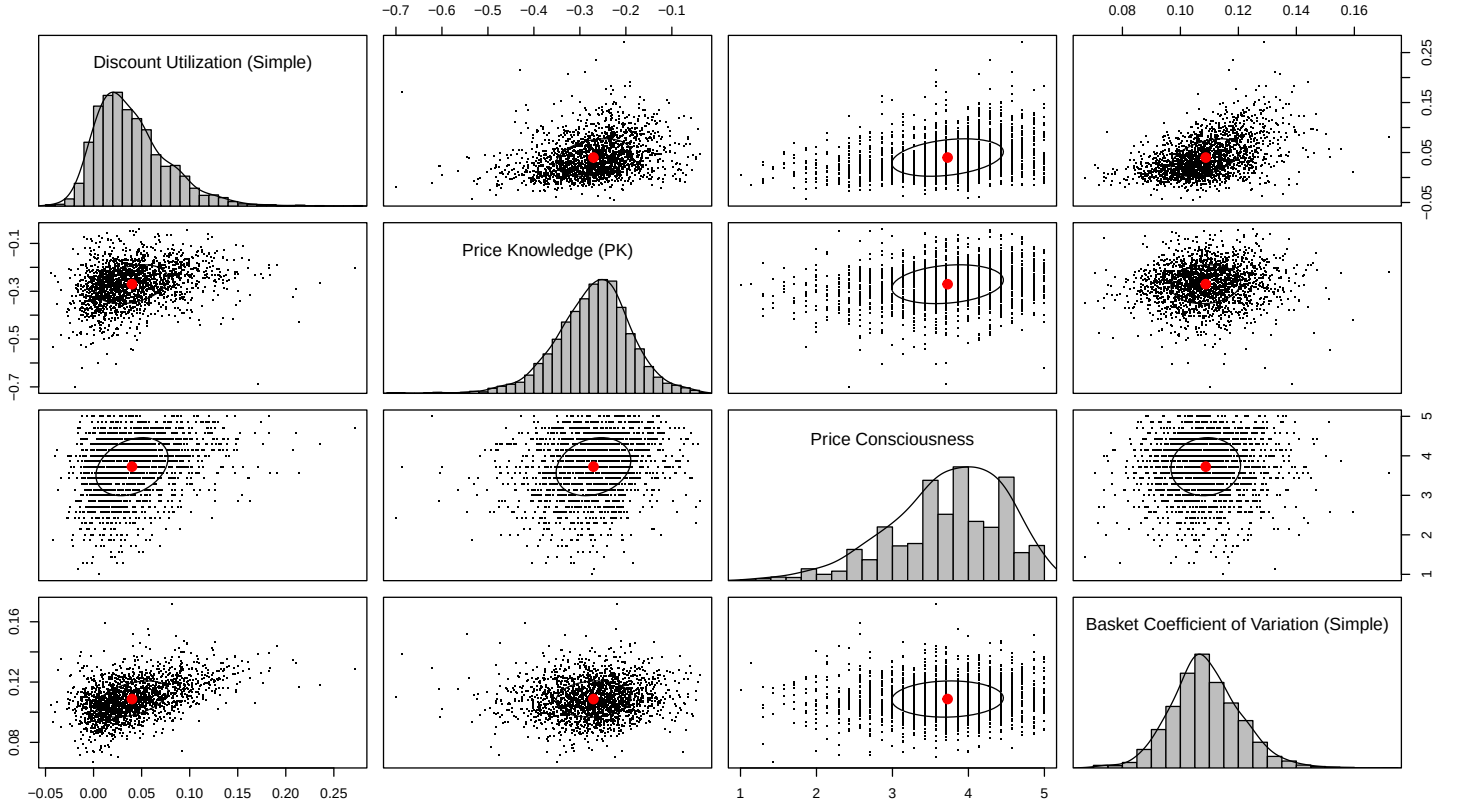


Figure C.3: Histograms, density plots, scatterplots, and correlation ellipses are presented for discount utilization, price knowledge, price consciousness, and the price variation in respondents' average shopping baskets. The basket coefficient of variation and discount utilization are calculated based on respondents' purchasing history within the categories covered by the survey. Discount utilization is defined as the percentage difference between the average price paid and the average shelf price per quarter, averaged over quarters and then over chain x product combinations. Price knowledge (PK) is measured as the average percentage absolute deviation between the respondent's recalled prices and the true prices. Price consciousness is calculated as a weighted average of Likert scale questions 1–7. The basket coefficient of variation represents the respondent's sales-share-weighted average of each chain x product combination's coefficient of variation, defined as the historical standard deviation of price divided by the average price.

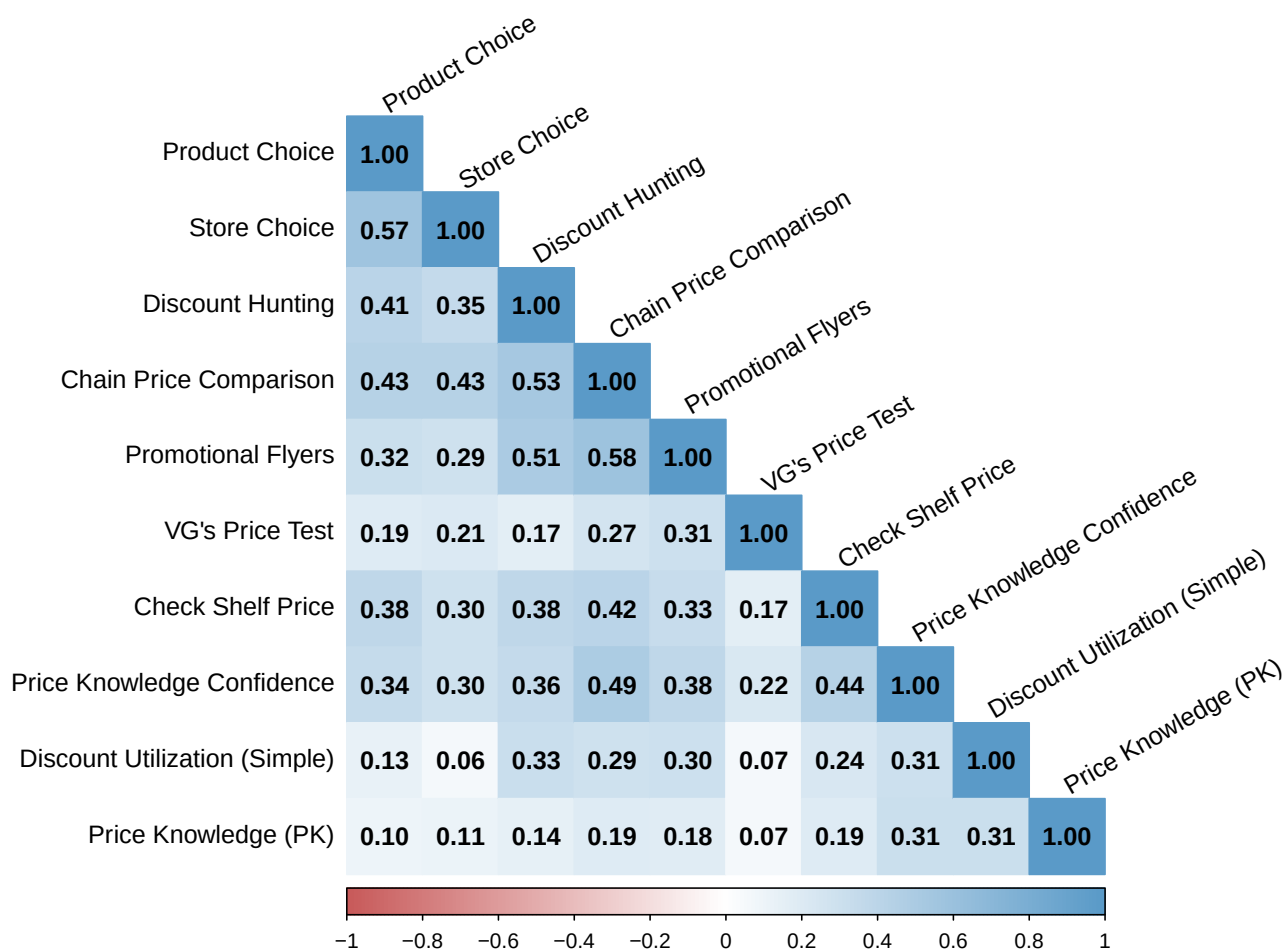


Figure C.4: Each cell contains the *Spearman's rank correlation coefficient* for Likert scale questions 1–8 and price knowledge. Price knowledge (PK) is measured as the average percentage absolute deviation between the respondent's recalled prices and the true prices. Price knowledge (20%) is the share of the respondent's recalled prices that fall within a 20% margin of error.

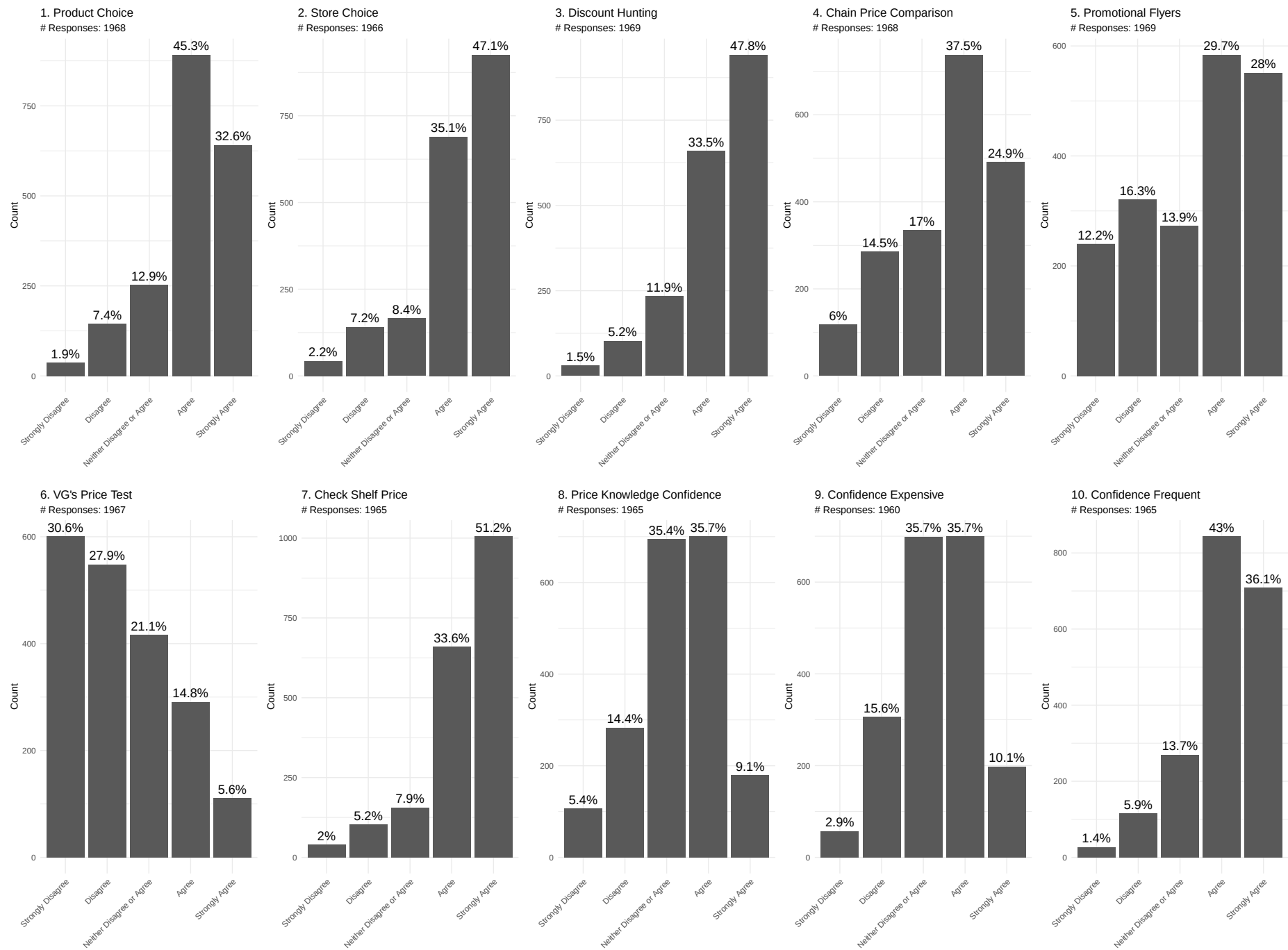


Figure C.5: This figure contains barcharts displaying the distribution of answers for each Likert scale question. The number of answers are given by the height of the bars, and the percentage share is indicated on top of the bars.

Table C.2: Variable Importance Measures for the Discount Utilization Regression

	Average R^2	First R^2	Last R^2	Standardized Coefficients
Basket Coefficient of Variation	0.168	0.204	0.135	0.393
Price Consciousness	0.063	0.089	0.041	0.226
Price Knowledge (PK)	0.052	0.081	0.032	0.188
Average Weekly Store Visits	0.019	0.029	0.011	0.148
3 CLP Memberships	0.012	0.024	0.004	0.071
Mainly Uses Discount Stores	0.006	0.006	0.001	-0.030
# Frequented Grocery Chains	0.005	0.015	0.000	0.005
Rural	0.004	0.004	0.003	-0.057
Average Weekly Variety	0.004	0.001	0.006	-0.108
Full-Time Employment	0.004	0.005	0.001	-0.044
Age: 40-59 Years	0.004	0.002	0.005	0.095
Age: Above 60 Years	0.002	0.000	0.004	0.101
Has Children	0.002	0.001	0.003	0.068
Main Grocery Shopper	0.002	0.005	0.000	0.023
Household Income > 950' NOK	0.002	0.003	0.000	-0.007
Male	0.001	0.002	0.000	0.013
Single Person Household	0.001	0.000	0.001	0.032
Higher Education	0.000	0.000	0.000	0.024

Notes: This table presents importance measures for the regression in Column 4 of Table 3.1. The variables are sorted by their average R^2 contributions across all regressor orderings (Column 1). Column 2 reports the R^2 from a simple regression using only the given variable, Column 3 shows the R^2 contribution of the variable when added to a model that includes all other variables, i.e. the "usefulness" of the variable. The standardized coefficients in Column 4 are obtained by rescaling the variables to have zero mean and unit variance before running the regression.

Table C.3: Variable Importance Measures for the Price Knowledge Regression

	Average R^2	First R^2	Last R^2	Standardized Coefficients
Category Size	0.043	0.050	0.039	-0.214
Has Purchased Product at Chain	0.013	0.021	0.006	0.122
Sales	0.006	0.012	0.003	0.062
Price Learning Effort	0.003	0.005	0.002	0.053
Price Coefficient of Variation	0.002	0.004	0.001	-0.028
Months Since Last Purchase	0.002	0.004	0.000	-0.014
Private Label	0.002	0.002	0.000	0.019
Supermarket Chain	0.002	0.001	0.002	0.064
Male	0.001	0.002	0.001	-0.029
3 CLP Memberships	0.001	0.002	0.000	0.020
Average Weekly Chain Variety	0.001	0.002	0.001	-0.056
Main Grocery Shopper	0.001	0.001	0.000	0.023
Price Sensitivity	0.001	0.002	0.000	0.001
Average Weekly Chain Visits	0.001	0.002	0.000	0.037
Higher Education	0.000	0.000	0.000	-0.020
# Frequented Grocery Chains	0.000	0.001	0.000	0.010
Age: 40-59 Years	0.000	0.000	0.000	0.012
Single Person Household	0.000	0.000	0.000	-0.001
Has Children	0.000	0.000	0.000	-0.001
Full-Time Employment	0.000	0.000	0.000	-0.008
Household Income > 950' NOK	0.000	0.000	0.000	0.016
Age: Above 60 Years	0.000	0.000	0.000	0.004
Mainly Uses Discount Stores	0.000	0.000	0.000	0.006
Rural	0.000	0.000	0.000	-0.003

Notes: This table presents importance measures for the regression in Column 3 of Table 4.1. The variables are sorted by their average R^2 contributions across all regressor orderings (Column 1). Column 2 reports the R^2 from a simple regression using only the given variable, Column 3 shows the R^2 contribution of the variable when added to a model that includes all other variables, i.e. the "usefulness" of the variable. The standardized coefficients in Column 4 are obtained by rescaling the variables to have zero mean and unit variance before running the regression.

Table C.4: Discount utilization regressions: sales weights

	Discount Utilization		
	(1)	(2)	(3)
Price Knowledge	0.070*** (0.010)	0.073*** (0.007)	0.054*** (0.007)
Male		0.000 (0.001)	0.000 (0.001)
Single Person Household		0.000 (0.002)	0.001 (0.002)
Has Children		-0.001 (0.001)	0.002 (0.001)
Full-Time Employment		-0.004* (0.002)	-0.002 (0.001)
Higher Education		-0.001 (0.001)	0.000 (0.001)
Rural		-0.006*** (0.001)	-0.003* (0.001)
Household Income > 950' NOK		0.000 (0.001)	0.001 (0.001)
Age: 40-59 Years		0.003* (0.001)	0.004** (0.001)
Age: Above 60 Years		0.006** (0.002)	0.005** (0.002)
Price Sensitivity		0.003** (0.001)	0.001 (0.001)
Price Learning Effort			0.005*** (0.001)
# Frequented Grocery Chains			0.000 (0.000)
Mainly uses discount stores			-0.008*** (0.002)
Average Weekly Store Visits			0.003*** (0.001)
Main Grocery Shopper			0.002 (0.001)
Average Weekly Variety (per 10 products)			-0.003*** (0.001)
Three Customer Program Memberships			0.003** (0.001)
Basket Coefficient of Variation			0.566*** (0.051)
Constant	0.040*** (0.003)	0.032*** (0.005)	-0.042*** (0.007)
Observations	2,028	1,935	1,928
Dependent variable mean	0.02	0.02	0.02
R ²	0.04	0.08	0.31

Notes: This table reports results from estimation of Equation ???. The dependent variable, individual-level discount utilization, is the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product $\times$ chain combinations, with the latter using sales weights. Price knowledge is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. The average weekly variety is calculated as the average number of unique products purchased per week over the sample period. Price sensitivity and learning effort are calculated as the factor-weighted averages of Likert-scale survey items 1–3 and 4–7, respectively. Basket CV is calculated as the coefficient of variation of shelf prices over the sample period, averaged across all products in the consumer's purchase history, using sales weights. Column 1 reports the simple regression on price knowledge, Column 2 adds controls for demographics and price sensitivity, and Column 3 adds controls for shopping habits and price learning effort. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

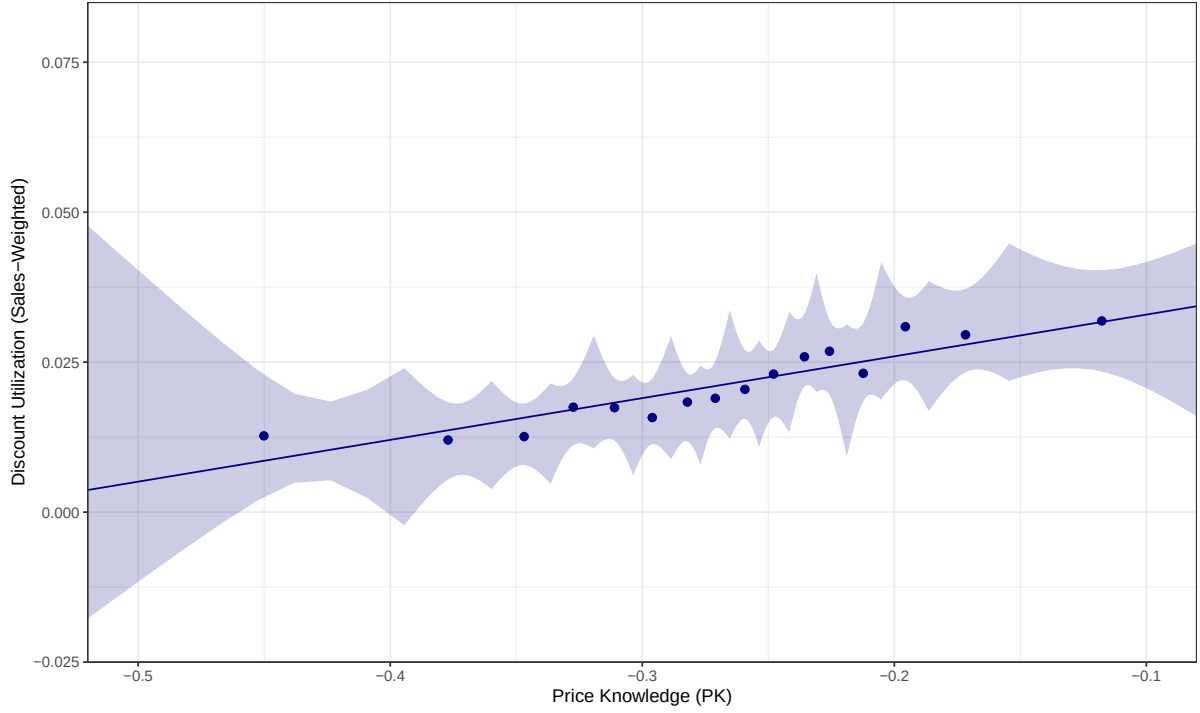


Figure C.6: Estimate of the expected value of discount utilization conditional on price knowledge using a binned scatterplot with a linear fit. The number of bins are chosen to minimize the integrated mean squared error. Confidence bands at 95 % are calculated using piecewise polynomials.

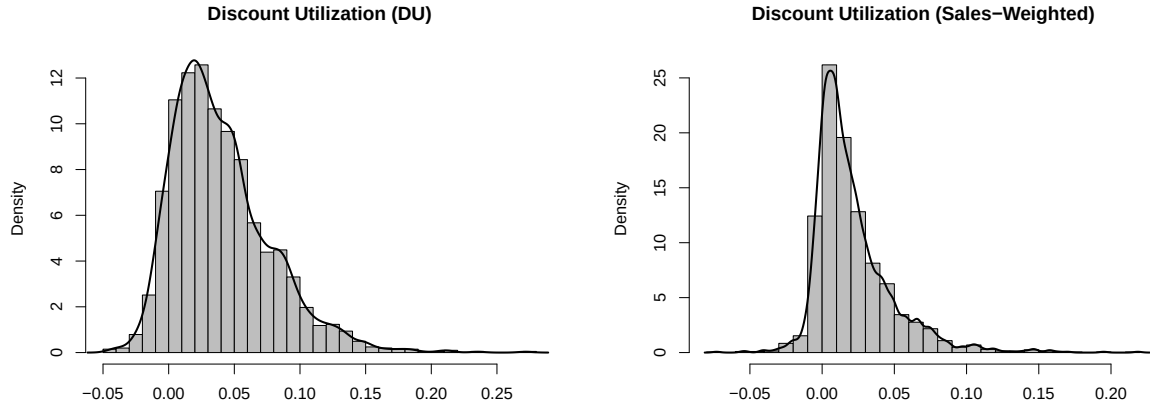


Figure C.7: Histograms and density estimates of discount utilization based on simple averages and sales-weighted averages across purchased products. Histogram bin sizes are chosen using Scott's rule and density bandwidths are chosen using Sheather-Jones' plug-in estimator.

D Full Consumer Survey

The survey used in this article was originally presented to respondents in Norwegian. Below, we present an English version of the complete survey, translated using Google Translate.

The survey was distributed via newsletter to Trumf members from the 4th to the 12th of June 2024. The median respondent spent 8.7 minutes completing the questionnaire. Of the 2505 respondents, 418 did not complete the survey in its entirety, yielding an attrition rate of 16.7%. Most of those who dropped out did so before the price recall section.

There were a few inconsistencies in the Likert scale response options. Two different Norwegian synonyms for "Strongly" ("Svært" and "Veldig") were used for the extreme response categories, and the adverb "Slightly" appeared only in the "Agree" option. It is difficult to determine whether this influenced respondents' choices, but if anything, we suspect it may have marginally shifted some answers between the two outer categories on either side.

Introduksjon

Price knowledge and purchasing behavior

You are hereby invited to participate in a research project led by [FOOD](#) , a collaboration between the Norwegian School of Economics (NHH) and NorgesGruppen. The study aims to explore the relationship between grocery customers' price knowledge and their purchasing behavior.

Project overview

Participation involves answering questions about demographics, shopping habits, and price awareness. No directly identifiable information about you will be shared with the FOOD project.

A selection of your purchase history from Trumf, which will be linked to the answers you provide in the survey, will be shared with the FOOD project. This includes information about selected products from product groups in the survey, sales amount, quantity/weight, chain and date of purchase, and number of purchases and average shopping cart size per week from January 1, 2023 to June 12, 2024.

Trumf will process the sales data and link it to your survey responses, but they will not have access to your responses.

Participation details

Participation in the survey is voluntary. The information is used only in the FOOD project and is treated confidentially in accordance with data protection regulations. The project will end in 2024, after which only anonymized data will be stored on a secure server at NHH, accessible to FOOD researchers.

Once you have completed the survey, you will be entered into the draw to win:

- Main prize: Trump bonus of NOK 5,000
- Partial prizes for 10 pieces: Trump bonus of 500 kr

The survey takes approximately **5-10 minutes** to complete.

By answering the survey, you agree to participate in the research project. If you do not wish to participate, please close the tab.

We greatly appreciate your participation!

Demografi

What is your age?

Under 30 years old

30-39 years old

40-49 years

50-59 years

60-69 years

70 years or over

What gender are you?

Woman

Man

Other:

How many members are there in your household?

1

2

3

4

5

6 or more

How many children under the age of 18 are there in your household?

0

1

2

3

4 or more

Choose the option that *best* describes your work situation:

Full-time employee

Part-time employee

Self-employed

Not at work

Student

Retired

Other:

What is your highest completed education?

Primary school

High school

Vocational school

University/college

Other:

What is your household's total annual income before taxes?

Total income includes all wage income, business income, capital income, pensions, and other taxable and tax-exempt transfers.

Under 300,000 NOK

300,000 - 600,000 kr

600,000 - 950,000 kr

950,000 - 1,350,000 kr

1,350,000 - 1,800,000 kr

Over 1,800,000 kr

How would you describe the area you live in?

City

Suburb

Rural

Handlevaner

Which of these grocery chains do you usually shop at during a typical month?

Check multiple boxes if applicable.

Kiwi

Rema 1000

Coop Extra

Menu

Spar/Eurospar

Coop Note

Coop Mega

Coop Price

Joker

Bottom price

Coop Market

The convenience store

The food corner

Oda

You did not check these chains. Which of them would it have been practical for you to shop at in your everyday life?

Check multiple boxes if applicable.

» Kiwi

» Rema 1000

» Coop Extra

» Menu

» Spar/Eurospar

» Coop Note

» Coop Mega

- » Coop Price
- » Joker
- » Bottom price
- » Coop Market
- » The convenience store
- » The food corner
- » Oda

Which of these chains do you shop at the most?

- » Kiwi
- » Rema 1000
- » Coop Extra
- » Menu
- » Spar/Eurospar
- » Coop Note
- » Coop Mega
- » Coop Price
- » Joker
- » Bottom price
- » Coop Market
- » The convenience store
- » The food corner
- » Oda

How many times do you shop for groceries (on average) per week?

- 0-1
- 2-3
- 4-5
- 6 or more

How many loyalty programs are you a member of (Æ, Trumf, Coop)?

- 1
- 2
- 3 or more

Are you primarily responsible for purchasing groceries for the household?

Yes

No

Shared responsibility

Priskunnskap - Intro

You will now be asked what you think the current shelf price is for 18 different grocery products.

Any offers or promotions should be taken into account, but you can disregard personal offers and coupons.

We ask that you answer as best you can based on your own memory.

Frokostprodukter

What is the current shelf price of **200g Wasa Rye Crackers** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **400g Nora homemade strawberry jam** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **110g sliced cooked ham from Gilde** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **1.1kg Bjørn easy-cooked oatmeal from AXA** in the following stores?



Number of kroner:

Kiwi

Menu

Kjøtt og fisk

What is the current shelf price of **400g of minced meat from First Price** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **600g Gilde grill sausages** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **4x125g salmon fillets from First Price** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **420g cod fillet from Findus** in the following stores?



Number of kroner:

Kiwi

Menu

Meieriprodukter

What is the current shelf price of **1l Tine low-fat milk 0.5% fat** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **12 breakfast eggs from Prior** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **500g Tine dairy butter** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **1kg of Norvegia yellow cheese** in the following stores?



Number of kroner:

Kiwi

Menu

Husholdningsprodukter og tørrvarer

What is the current shelf price of **2kg of Møllerens sifted wheat flour** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **500ml of Zalo Ultra dishwashing detergent** in the following stores?



Kiwi

Menu

Number of kroner:

What is the current shelf price of **1kg of spaghetti from Barilla** in the following stores?



Kiwi

Menu

Number of kroner:

What is the current shelf price of **250g Evergood Classic filter ground coffee** in the following stores?



Kiwi

Menu

Number of kroner:

Brus, snacks og søtsaker

What is the current shelf price of **200g Freia milk chocolate** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **1.5l Pepsi Max** in the following stores (without deposit)?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **195g Sørlandschips sea salt** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **250ml Bendit smoothie with raspberries and strawberries** in the following stores?



Number of kroner:

Kiwi

Menu

Frukt og grønt

What is the current shelf price of a **Norwegian cucumber from Gartner** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price for **2 packs of ripe avocados from Bama** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **1kg of bananas from Bama** in the following stores?



Number of kroner:

Kiwi

Menu

What is the current shelf price of **1kg of potatoes in bulk** in the following stores?



Number of kroner:

Kiwi

Menu

Prisbevissthet - 1-by-1

Finally, we ask you to indicate the extent to which you agree with 10 different statements.

Price is crucial for my choice of *products* .

Strongly disagree

Disagree

Neither agree nor
disagree

Slightly agree

Strongly agree

Price is crucial for my choice of *store* .

Strongly disagree

Disagree

Neither agree nor
disagree

Slightly agree

Strongly agree

I buy on offers and promotions as often as I can.

Strongly disagree	Disagree	Neither agree nor disagree	Slightly agree	Strongly agree
I usually compare prices between chains.				
Strongly disagree	Disagree	Neither agree nor disagree	Slightly agree	Strongly agree
I usually check the grocery chains' customer flyers for prices.				
Strongly disagree	Disagree	Neither agree nor disagree	Slightly agree	Strongly agree
I usually check the results in VG's price comparison test.				
Strongly disagree	Disagree	Neither agree nor disagree	Slightly agree	Strongly agree
I usually check the shelf price of the product before I buy it.				
Strongly disagree	Disagree	Neither agree nor disagree	Slightly agree	Strongly agree
I have good knowledge of grocery prices.				
Strongly disagree	Disagree	Neither agree nor disagree	Slightly agree	Strongly agree
I have better knowledge about the prices of groceries I buy frequently.				
Strongly disagree	Disagree	Neither agree nor disagree	Slightly agree	Strongly agree
I have better knowledge about the prices of groceries that cost a lot.				
Strongly disagree	Disagree	Neither agree nor disagree	Slightly agree	Strongly agree